GEOGRAPHY • RAS PRELIMS NOTES

Geography of Rajasthan

Physiography, rivers & lakes, climate, forests & soils, agriculture, livestock, irrigation, population, tribes, minerals and tourism - the highest-weight section in RAS

Inside these notes

- District-wise tables for facts RAS loves to match and order
- Census 2011 orders, river systems and mineral belts, fully covered
- Memory hooks and exam traps in every chapter

Dinesh Jangir
Founder, StuDy
Notes made for RAS aspirants - simple language, connected concepts, PYQ-based facts.

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How to use: read a chapter fully once, then revise only the tables, MEMORY HOOKS and EXAM TRAPS. Every fact here is written the way RAS papers ask it.

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StuDy

Geography of Rajasthan: Location, Extent and Physiography

1. Location and Extent

Rajasthan is the largest state of India. It lies in the north-west of the country, between the Indo-Gangetic plain in the east and the Thar desert in the west. Think of the state as a tilted kite (a rough rhombus) whose western side faces Pakistan and whose south-eastern corner touches the Vindhyan country of Madhya Pradesh.

- **Latitudinal extent:** 23°03' N to 30°12' N (about 826 km from north to south)
- **Longitudinal extent:** 69°30' E to 78°17' E (about 869 km from west to east)
- **Total area:** 3,42,239 sq km, which is **10.4% of India's area** — the largest of all states
- **Rank in India:** 1st by area; 7th by population
- **Capital:** Jaipur

The **Tropic of Cancer (23°30' N)** passes through the southern tip of the state, through **Banswara and Dungarpur** districts. Banswara town lies almost exactly on the Tropic. This is why southern Rajasthan is hotter and wetter than the rest of the state.

MEMORY HOOK: "23 to 30 latitude, 69 to 78 longitude, 3.42 lakh area" — chant this one line; half the location questions of RAS end here.

EXAM TRAP: Rajasthan became the largest state only after Madhya Pradesh was divided (1 November 2000, Chhattisgarh separated). Before that, Madhya Pradesh was bigger.

PYQ PATTERN: RAS repeatedly asks the districts at the four corners of the state and the length of the Pakistan border — learn both.

Borders — who touches Rajasthan?

Rajasthan touches **one country (Pakistan) and five states** (Punjab, Haryana, Uttar Pradesh, Madhya Pradesh, Gujarat). The total land boundary is about 5,920 km, of which 1,070 km is international.

Neighbour	Border length	Main bordering districts of Rajasthan
Pakistan	1,070 km	Sriganganagar, Bikaner, Jaisalmer, Barmer
Madhya Pradesh	1,600 km (longest state border)	Banswara, Pratapgarh, Chittorgarh, Kota, Baran, Jhalawar, Dholpur, Karauli, Bhilwara, Udaipur
Haryana	1,262 km	Hanumangarh, Churu, Jhunjhunu, Sikar, Alwar, Bharatpur
Gujarat	1,022 km	Banswara, Dungarpur, Udaipur, Sirohi, Jalore, Barmer
Uttar Pradesh	877 km	Bharatpur, Dholpur
Punjab	89 km (shortest state border)	Sriganganagar, Hanumangarh

EXAM TRAP: Madhya Pradesh gives the LONGEST inter-state border (1,600 km); Punjab gives the SHORTEST (89 km). Pakistan border (1,070 km) is a separate figure — do not mix it with the state borders.

MEMORY HOOK: The only state touching both Punjab and Uttar Pradesh sides of Rajasthan is Haryana in between — draw the map clockwise: Pakistan, Punjab, Haryana, UP, MP, Gujarat.

Extreme districts and size extremes

Direction / fact	District
Northernmost district	Sriganganagar
Southernmost district	Banswara
Westernmost district	Jaisalmer
Easternmost district	Dholpur
Largest district by area	Jaisalmer (38,401 sq km)
Smallest district by area	Dholpur (3,034 sq km)
District touching the maximum number of other districts	Pali (touches 8 districts)

Eight districts of Rajasthan touch **no other state and no foreign country** — they are fully surrounded by Rajasthan itself: **Jodhpur, Pali, Nagaur, Ajmer, Rajsamand, Tonk, Bundi and Dausa**. Every other district shares a border with a neighbouring state or Pakistan.

PYQ PATTERN: "Which district does not share a border with any other state?" — the answer is always one of these eight.

MEMORY HOOK: Jaisalmer is so large that it alone is bigger than states like Kerala's two-thirds; Dholpur, the smallest, is less than one-twelfth of it.

Note: District names and Census figures in these notes follow the classic 33-district map of Census 2011, which is the frame RAS papers use. For administration the state is grouped into 7 divisions: Jaipur, Jodhpur, Bikaner, Ajmer, Udaipur, Kota and Bharatpur.

2. The Four Physiographic Divisions — the skeleton of all Rajasthan geography

Almost every Rajasthan geography question stands on one map: the state divides into **four physical divisions**. The Aravalli range runs diagonally (south-west to north-east) and cuts the state into two unequal halves — the dry west (about 61%) and the wetter east (about 39%). Memorise the four divisions with their shares first; every river, soil, crop and mineral later in these notes hangs on this frame.

Division	Share of state area	General character
1. Western Sandy Plain (Thar Desert)	about 61%	Sand dunes, very low rainfall, internal drainage
2. Aravalli Range and Hill Region	about 9%	Oldest fold hills, the state's backbone and water divide
3. Eastern Plain	about 23%	River plains of Chambal, Banas, Banganga and Mahi
4. South-Eastern Plateau (Hadoti Plateau)	about 7%	Vindhyan plateau, black soil, Chambal valley

3. Division 1 — The Western Sandy Plain (Thar Desert)

The Thar is one of the most densely populated deserts of the world. About **62% of the Thar desert lies in Rajasthan** (roughly 85% of the whole Thar is within India). In Rajasthan it covers twelve districts fully or partly: Jaisalmer, Barmer, Bikaner, Jodhpur, Jalore, Pali, Nagaur, Churu, Sikar, Jhunjhunu, Hanumangarh and Sriganganagar. Its eastern limit is roughly the Aravalli hills and the **50 cm rainfall line (isohyet)**.

Physiographers split this plain into two parts:

(a) The Great Indian Desert or Marusthali (proper dune country)

This is the true sandy desert west and north-west, full of shifting and fixed dunes. In its south-west (Jaisalmer, Barmer, south Bikaner) the surface is rocky in patches — ancient granite and rhyolite outcrops near the surface form **hammadas** (rocky desert floors), and gravelly plains between dunes are called **reg**. Flat salt-encrusted depressions that fill briefly after rain are **playas**, locally called **ranns** or **dhands**.

(b) The Rajasthan Bangar (semi-arid transitional plain)

East of the dune country, up to the Aravalli foot, lies a semi-arid plain of older alluvium and gravel — the Bangar. RAS questions love its four sub-regions:

Sub-region of Rajasthan Bangar	Where	Key identity
Ghaggar Plain	Sriganganagar, Hanumangarh (north)	Flat alluvial plain of the Ghaggar; canal-irrigated granary
Shekhawati Region	Sikar, Jhunjhunu, part of Churu (north-east)	Sandy plain of the Kantli basin; painted havelis
Nagaur Upland	Nagaur and around (centre)	Slightly raised tract with salt depressions — Sambhar, Didwana, Kuchaman
Luni Basin (Godwar)	Pali, Jalore, Sirohi foothills, east Barmer (south)	Plain drained by the Luni and its tributaries up to the Rann

PYQ PATTERN: In RAS 2021 the paper directly asked this matching — **Luni Basin and Shekhawati are sub-regions of the Rajasthan Bangar**, not of the Marusthali. When in doubt, remember: Bangar = the four named sub-regions (Ghaggar, Shekhawati, Nagauri upland, Luni basin).

MEMORY HOOK: "Bangar has four children — Ghaggar in the north, Shekhawati in the north-east, Nagaur in the middle, Luni in the south."

Sand dunes — the forms the exam asks

Dune type	Shape / fact
Longitudinal (seif) dunes	Long ridges parallel to the wind; the MOST common type in the Thar
Transverse dunes	Ridges across the wind direction
Barchan (crescentic) dunes	Crescent-shaped, horns pointing downwind; the most mobile and most harmful to farms and roads
Parabolic dunes	Hairpin/U-shaped, like a barchan reversed; fixed partly by vegetation
Star dunes	Pyramid-shaped with several arms; found around Mohangarh (Jaisalmer)

Dune type	Shape / fact
Shrub-coppice dunes (nebkha)	Small mounds collected around bushes

EXAM TRAP: **Barchans** are the most destructive/mobile; **longitudinal** dunes are the most numerous. The exam swaps these two adjectives deliberately.

Wind erosion in this plain is the state's biggest soil problem. Villages protect fields with the hardy desert grass **sewan** and with shelter belts of khejri.

4. Division 2 — The Aravalli Range and Hill Region

The Aravalli is the **oldest fold mountain system of India and one of the oldest in the world**, formed in the Precambrian era (Proterozoic, roughly 1.8–2.5 billion years ago) — far older than the Himalaya. It is a much-worn, residual range: what we see today is the stump of once-lofty mountains.

- **Total length:** about 692 km, from Palanpur (Gujarat) to the Delhi Ridge
- **Length in Rajasthan:** about 550 km (roughly 80%), running **south-west to north-east** from Khed Brahma/Sirohi side up through Udaipur, Ajmer and Jaipur to Alwar
- **Highest peak: Guru Shikhar (1,722 m)** on Mount Abu, Sirohi — the highest point between the Himalaya and the Nilgiris
- **Role:** It is the **water divide** of Rajasthan. Rivers east of it (Banas, Chambal system) reach the Bay of Bengal; rivers west of it (Luni) reach the Arabian Sea or die inland. It also checks the spread of the desert towards the east.

Because the range runs **parallel** to the direction of the Arabian Sea monsoon winds, those winds pass along it without being lifted, so the west stays dry — this single line explains the rainfall map of the whole state.

Three parts of the Aravalli in Rajasthan

Part	Districts	Character
Southern Aravalli (Mewar hills)	Sirohi, Udaipur, Rajsamand, Dungarpur	Highest and most continuous; Abu block, Girwa hills, Bhorat plateau
Central Aravalli (Merwara hills)	Ajmer, Rajsamand, Pali	Narrower, broken hills; Desuri and other passes; source of the Luni
North-Eastern Aravalli	Jaipur, Alwar, Sikar, Jhunjhunu, Dausa	Low, scattered hills (Bairath, Khandela, Harsh hills); copper belt of Khetri

The **Abu block** in the south-west stands slightly apart from the main range, separated by the valley of the West Banas river. Around Udaipur city, saucer-shaped hills enclose a basin — this cup-like tract is the **Girwa**, and its hilly rim country is the **Magra**. Between Kumbhalgarh (Rajsamand) and Gogunda (Udaipur) spreads the high **Bhorat plateau**. East of Udaipur towards Salumbar lies the **Lasadia plateau**.

Important passes (locally called **nal**) through the range: **Desuri ki Nal** (links Marwar/Pali with Mewar/Udaipur), **Kamli Ghat** and **Goram Ghat** (on the Marwar–Mewar railway route) and **Hathi Guda ki Nal**.

Major peaks

Peak	Height	District
Guru Shikhar	1,722 m (highest in Rajasthan)	Sirohi (Mount Abu)

Peak	Height	District
Ser	1,597 m	Sirohi
Dilwara	1,442 m	Sirohi
Jarga	1,431 m	Udaipur
Achalgarh	about 1,380 m	Sirohi
Raghunathgarh	1,055 m (highest of the north-east Aravalli)	Sikar
Sajjangarh	about 940 m	Udaipur
Taragarh	873 m	Ajmer

MEMORY HOOK: Peaks descend in pairs from Abu — Guru Shikhar, Ser, Dilwara all sit in Sirohi; only Jarga breaks the pattern in Udaipur.

EXAM TRAP: The Aravalli is a **fold** mountain (old); the Vindhyas to the south-east are largely block/escarpment country. Do not call the Aravalli a young or volcanic range.

Also remember the **Malani hills** of Barmer–Jalore: ancient volcanic (rhyolite) hills of the south-western plain, among the largest rhyolite outcrops in the world; the granitic ring hills near Siwana (Barmer) are called the **Chhappan hills** — do not confuse them with the Chhappan plain of Banswara in the east.

5. Division 3 — The Eastern Plain

East of the Aravalli the land opens into plains built by rivers flowing towards the Yamuna–Ganga system. Rainfall is higher (50–90 cm), soils are alluvial and farming is intensive. It has four sub-plains, each tied to one river:

- **Banas Plain:** The broad level tract of Tonk, Sawai Madhopur, Dausa, Bhilwara and Ajmer side, drained by the Banas and its tributaries (Bedach, Kothari, Khari, Morel). Bhilwara–Tonk part of it is locally upland-ish towards the **Upar Mal** plateau rim near Bundi.
- **Banganga Plain:** Jaipur–Dausa–Bharatpur tract drained by the Banganga and Gambhir; rich alluvium, dense villages.
- **Chambal Basin (the ravine country):** In Dholpur, Karauli and Sawai Madhopur the Chambal and its tributaries have cut deep gullies into the alluvium, creating badlands called **beehad**. The rugged Karauli–Dholpur tract of ravines and table-topped hills is the **Dang** region.
- **Mahi Plain (Vagad):** Banswara and Pratapgarh in the far south, drained by the Mahi, Som and Jakham. The flood plain between the Mahi and its tributaries near Banswara–Pratapgarh is the **Chhappan plain** (the tract of "fifty-six" stream-cut villages), and the Mahi bank country of Pratapgarh is called **Kantal**. This is the wettest, greenest corner of the state.

PYQ PATTERN: River–plain matching is a favourite: Banas plain, Banganga plain, Chambal ravines and Mahi plain are asked with their rivers and districts in one table.

6. Division 4 — The South-Eastern Plateau (Hadoti Plateau)

The south-east corner — **Kota, Bundi, Baran, Jhalawar** and the eastern edge of Chittorgarh — is the Rajasthan share of the great Vindhyan plateau of central India. It is a land of black cotton soil, sandstone scarps and the deep Chambal valley. Its parts:

- **Vindhyan scarplands** along the Chambal in Kota–Dholpur direction

- **Shahbad Upland** (Baran) — the highest part of Hadoti
- **Dag–Gangdhar Upland** (south-west Jhalawar)
- **Jhalawar Plateau** in the south, sloping to Madhya Pradesh
- **Mukundra Hills** — a long crescent-shaped range running across Kota and Jhalawar, now a tiger reserve
- **Bundi Hills** with Taragarh fort

A famous curiosity of this plateau: the **Ramgarh crater** near Ramgarh (Baran) — a circular structure about 3.5 km wide formed by a meteorite impact, now inside the Ramgarh Vishdhari wildlife area.

HOOK for the whole physiography chapter: **West is sand, middle is stone, east is soil, south-east is black plateau** — 61-9-23-7.

PYQ PATTERN: RAS asks the physiographic region of a district directly (for example, "Shahbad upland lies in which district?" — Baran). Learn one landmark per sub-region and the matching becomes free marks.

7. Plateaus and landmark landforms at a glance

Rajasthan GK names its small plateaus and landforms individually, and RAS matches them to districts. Keep this ready list:

Landform	District / region	Identity
Arbuda (Oriya) plateau	Sirohi	The high Abu tableland that carries Guru Shikhar — the highest plateau surface of the state
Bhorat plateau	Udaipur–Rajsamand	High plateau between Kumbhalgarh and Gogunda
Lasadia plateau	Salumbar–Udaipur (east)	Dissected plateau of the southern Aravalli edge
Mesa plateau	Chittorgarh	Flat-topped tableland on which Chittorgarh fort stands
Upar Mal plateau	Bhilwara–Bundi	Raised tract between the Banas plain and Hadoti
Shahbad upland	Baran	Highest part of the Hadoti plateau
Dag–Gangdhar upland	Jhalawar	South-western rim of Hadoti
Girwa	Udaipur	Saucer-shaped ring of hills around the Udaipur basin
Magra	Udaipur–Rajsamand	Hilly rim country of the Girwa
Chhappan plain	Banswara–Pratapgarh	Mahi flood plain of "fifty-six" villages
Kantal	Pratapgarh	Mahi-bank tract
Dang	Karauli–Dholpur	Ravine-cut badland along the Chambal
Godwar	Pali–Jalore–Sirohi foothills	The Luni basin tract (in the Bangar)
Torawati	Sikar–Jhunjhunu	The Kantli valley country of Shekhawati
Vagad	Banswara–Dungarpur	Mahi-drained tribal south

Landform	District / region	Identity
Mewar, Marwar, Dhundhar, Hadoti, Shekhawati, Matsya	Udaipur side, Jodhpur side, Jaipur side, Kota side, Sikar side, Alwar–Bharatpur side	The historic-cultural regions every map question silently uses

MEMORY HOOK: When a question uses a regional name instead of a district — Vagad, Godwar, Dang, Kantal, Torawati — it is testing this very table. Translate the region into districts first, then answer.

Rivers and Lakes of Rajasthan

1. The three drainage systems

Rajasthan's rivers fall into three families. Just remember the Aravalli water divide and the families arrange themselves:

Drainage system	Share of drainage area (approx.)	Rivers
Bay of Bengal system	about 24%	Chambal and its tributaries (Banas, Kali Sindh, Parbati), Banganga, Gambhir
Arabian Sea system	about 18%	Luni, Mahi, Sabarmati, West Banas
Internal (inland) drainage	about 58% — the largest	Ghaggar, Kantli, Sabi, Kakni, Mentha, Ruparail and desert streams that die in sand or salt lakes

More than half of Rajasthan drains **nowhere** — its streams fade into sand or end in salt lakes like Sambhar. That one fact is the key to the state's water problem.

2. The Chambal — the state's only great perennial river

- **Origin:** Janapav hills near Mhow (Indore district, Madhya Pradesh), in the Vindhyas
- **Length:** about 1,051 km — the longest river connected with Rajasthan; about 376 km of its course touches/flows in the state
- **Course in Rajasthan:** Enters near Bhainsrorgarh (Chittorgarh), flows past Kota, then forms the boundary through Sawai Madhopur, Karauli and Dholpur before joining the Yamuna in Uttar Pradesh (near Etawah)
- **Ancient name:** Charmanwati
- **Character:** The only **perennial** river of Rajasthan (it flows all year). It has cut deep **ravines (beehad)** in its lower course — a fingerprint PYQ identity.
- **Projects:** Gandhi Sagar, Rana Pratap Sagar, Jawahar Sagar and Kota Barrage (details in the Irrigation chapter); the tri-state **National Chambal Sanctuary** protects gharial, mugger and the Gangetic dolphin along it.

Main tributaries joining in or near Rajasthan: **Banas** (its longest tributary), **Kali Sindh, Parbati, Parwan, Mej, Kuno and Alnia**.

Kali Sindh and Parbati (Hadoti pair)

- **Kali Sindh:** Rises near Bagli (Dewas, MP), enters Jhalawar, flows through Jhalawar and Kota and joins the Chambal. The **Ahu** is its tributary (Gagron fort stands at the Ahu–Kali Sindh confluence). The Kalisindh thermal power plant (Jhalawar) uses its water.
- **Parwan:** Rises in MP, flows through Jhalawar (Aklera side) and joins the Kali Sindh.

- **Parbati:** Rises in the Vindhyas of MP, flows through Baran (Kishanganj, Shahbad) and joins the Chambal in Kota district.
- **Mej:** Rises near Mandalgarh (Bhilwara), flows through Bundi and joins the Chambal.

EXAM TRAP: There are two different Parbatis in Indian geography questions; Rajasthan's Parbati belongs to the **Chambal** system and flows through **Baran**, not to the Luni or Banas.

3. The Banas — the longest river flowing entirely within Rajasthan

The Banas is the hero river of Mewar and the exact pattern RAS set in a recent paper, so learn it as a story:

- **Origin: Khamnor hills near Veron ka Math, close to Kumbhalgarh fort (Rajsamand)**, in the Aravalli
- **Length:** about 512 km — every kilometre of it lies **inside Rajasthan**, making it the state's longest wholly-internal river
- **Course:** Rajsamand → Udaipur side → Bhilwara → Tonk → Sawai Madhopur, joining the **Chambal near Rameshwar (Sawai Madhopur)**
- **Nick-name:** "Van ki Asha" — the hope of the forest
- **Chief project: Bisalpur dam** (Tonk) — the state's largest drinking-water project, supplying Jaipur

PYQ PATTERN: RAS asked: "The river that originates in the Kumbhalgarh hills, flows through the Mewar plain and has Bedach and Kothari as tributaries" — the answer is **Banas**. This single description is a ready-made PYQ; fix it permanently.

Tributaries of the Banas

Tributary	Origin / course	Note
Bedach	Gogunda hills (Udaipur); its upper stretch in Udaipur is the Ahar river	Flows through Udaipur and Chittorgarh, joins Banas near Gulabpura side
Kothari	Diver hills (Rajsamand)	Flows through Bhilwara; Meja dam stands on it
Khari	Hills of Rajsamand	Joins Banas near Deoli (Tonk)
Morel	Jaipur–Dausa hills	Joins Banas in Sawai Madhopur; Morel dam on it
Mashi	Kishangarh side (Ajmer)	Small tributary joining near Tonk side
Dai	Ajmer side	Joins Banas in Tonk

MEMORY HOOK: "Be-Ko-Kha-Mo" — Bedach, Kothari, Khari, Morel carry the Banas to the Chambal.

4. The Luni — the river of the desert (Marwar)

- **Origin: Nag Pahar (Nag hills) near Ajmer**, on the western slope of the Aravalli; near Govindgarh it is joined by the Saraswati stream coming from Pushkar, after which it takes the name Luni
- **Length:** about 495 km

- **Course:** Ajmer → Nagaur → Pali → Jodhpur → Barmer → Jalore → dies in the **Rann of Kutch** (Gujarat)
- **Water character:** Sweet (fresh) up to **Balotra (Barmer)**, saline after that — hence the name Luni, from "Lavanavari", the salt river. Its whole basin between Aravalli and the Rann is the **Luni Basin or Godwar region** of the Bangar.
- **Tributaries:** Jawai, Sukri, Bandi, Sagi, Lilri, Mithri, Guhiya, Jojari. The **Jawai** (rising in the Pali hills) is the largest; the Jawai dam (Pali) is the biggest dam of western Rajasthan. The **Bandi** is infamous for dyeing-and-printing pollution around Pali.

MEMORY HOOK: "Luni is sweet till Balotra" — the exam asks exactly where the Luni turns saline; the answer is Balotra.

EXAM TRAP: The Luni does NOT reach the open Arabian Sea; it is lost in the marshes of the **Rann of Kutch**.

5. The Mahi — the Ganga of Vagad

- **Origin:** Vindhyan hills of Madhya Pradesh (near Bhopawar, Dhar district)
- **Length:** about 583 km
- **Course:** Enters Rajasthan at **Khandu (Banswara)**, flows through Banswara, Dungarpur and Pratapgarh country (the **Vagad** region), then into Gujarat to the **Gulf of Khambhat**
- **Unique record:** The Mahi is the only river of India that **crosses the Tropic of Cancer twice**
- **Tributaries: Som, Jakham, Anas, Chap and Moren**. At **Beneshwar (Dungarpur)** the Som and Jakham join the Mahi — the tribal Triveni sangam
- **Projects: Mahi Bajaj Sagar** (Banswara, joint Rajasthan–Gujarat project) and Kadana dam (Gujarat)
- Tribals call it the **Ganga of Vagad**.

6. The Sabarmati and the West Banas

- **Sabarmati:** Rises in the Aravalli of **Udaipur district (near Padarla village, Jhadol side)**. Only about 45 km of its 371 km run lies in Rajasthan; the rest is Gujarat (Ahmedabad stands on it). Its main tributary from Rajasthan is the **Wakal** (rising near Gogunda, Udaipur); the Sei also joins the system.
- **West Banas:** Rises in the Sirohi Aravalli, flows south-west through Gujarat's Banaskantha (Deesa stands on it) to the Little Rann of Kutch.

EXAM TRAP: **Banas (east, → Chambal)** and **West Banas (Sirohi, → Rann of Kutch)** are two completely different rivers. RAS sets this confusion almost every cycle.

7. Banganga and Gambhir — the eastern pair

- **Banganga:** Rises in the **Bairath (Viratnagar) hills** of Jaipur country, flows through Jaipur, Dausa and Bharatpur, and finally joins the Gambhir near Rupbas; the combined waters reach the Yamuna in UP. Ramgarh dam (Jaipur) stands on it. Keoladeo's wetlands historically received its waters.
- **Gambhir (Utangan):** Rises in the Sapotra hills (Karauli), flows through Karauli and Bharatpur (the **Ajan Bund** on it feeds Keoladeo National Park), and joins the Yamuna near Agra. **Panchna dam** (Karauli) — the state's largest earthen dam — is built where five streams of this system meet.

8. Internal drainage — rivers that never reach the sea

- **Ghaggar:** Rises in the Shivaliks (Himachal), enters Rajasthan at **Hanumangarh** (Talwara lake is formed on it), flows past Suratgarh and dies in the sands near Anupgarh. It is called the **"Dead River"** and is identified with the lost Vedic **Saraswati**; the Harappan site **Kalibangan** stands on its bank. Its plain is the fertile Ghaggar plain of the Bangar.
- **Kantli:** Rises in the **Khandela hills (Sikar)**, flows about 100 km through Sikar and Jhunjhunu and is lost in sand near the Churu border. It is the longest purely internal-drainage stream of the Shekhawati region.
- **Sabi (Sahibi):** Rises in the Sewari hills (Jaipur), flows through Alwar into Haryana (towards the Najafgarh drain).
- **Kakni (Kakneya):** Rises in the Kotri hills (Jaisalmer); at about 17 km it is the **shortest river** of Rajasthan; it forms the sweet-water Bujh lake.
- **Ruparail:** Rises in the Udainath hills (Thanagazi, Alwar), flows to Bharatpur's marshes.
- **Mentha and Rupnagarh:** Small streams of Nagaur–Ajmer that end in **Sambhar Lake** (with the Khari of Ajmer and Khandela stream) — Sambhar is fed by four streams: Mentha, Rupnagarh, Khari and Khandela.

Master river table (for match-the-following practice)

River	Origin	Ends in / joins	Length	One-line identity
Chambal	Janapav hills, Mhow (MP)	Yamuna (UP)	1,051 km	Only perennial river; ravines
Banas	Kumbhalgarh hills (Rajsamand)	Chambal	512 km	Longest river wholly in Rajasthan
Luni	Nag Pahar, Ajmer	Rann of Kutch	495 km	Saline after Balotra
Mahi	Vindhyas (MP)	Gulf of Khambhat	583 km	Crosses Tropic of Cancer twice
Sabarmati	Aravalli, Udaipur	Gulf of Khambhat	371 km	Only 45 km in Rajasthan
Banganga	Bairath hills	Gambhir → Yamuna	—	Ramgarh dam on it
Ghaggar	Shivaliks	Lost in sand (Hanumangarh side)	—	"Dead river", Kalibangan on bank
Kantli	Khandela hills, Sikar	Lost in sand	100 km	Shekhawati's internal river
Kakni	Kotri hills, Jaisalmer	Bujh lake	17 km	Shortest river of the state

9. Lakes — salt lakes of the west, sweet lakes of Mewar

Rajasthan's lakes divide neatly: **salt lakes in the dry west and centre** (relics of an ancient sea, the Tethys) and **freshwater lakes in the south (Mewar)**, most of them artificial, built by the Maharanas.

Salt lakes

Lake	District	Fact
Sambhar	Jaipur–Ajmer–Nagaur border	Largest inland salt lake of India (about 230 sq km at full spread); supplies about 8.7% of India's salt; a **Ramsar site (1990)**; Sambhar Salts Ltd works it
Didwana	Nagaur (Didwana-Kuchaman)	Salt lake; its brine suits sodium-sulphate making
Pachpadra	Barmer	Famous for the best-quality salt; Kharwal community traditionally makes it
Lunkaransar	Bikaner	Northern salt lake of the desert
Phalodi, Kuchaman, Pokhran	Phalodi, Nagaur, Jaisalmer	Smaller salt pans

MEMORY HOOK: Sambhar's four feeder streams spell "MRKK" — Mentha, Rupnagarh, Khari, Khandela. If a question asks which river ends in Sambhar, pick from these four only.

Freshwater and famous lakes

Lake	District	Fact
Jaisamand (Dhebar)	Udaipur (Salumbar side)	Built by Maharana Jai Singh (1685–91) across the Gomati river; among the largest artificial lakes of Asia; islands Baba ka Magra (biggest) and Pyari (smallest)
Rajsamand	Rajsamand	Built by Maharana Raj Singh (1662–76) on the Gomati; at its Nau Chowki embankment the **Raj Prashasti**, one of the longest Sanskrit inscriptions, is carved on marble slabs
Pichola	Udaipur	Holds Jag Mandir and Jag Niwas (Lake Palace); linked to Fateh Sagar by Swaroop Sagar
Fateh Sagar	Udaipur	Rebuilt by Maharana Fateh Singh; Nehru Park island and a solar observatory stand in it
Udai Sagar, Badi, Swaroop Sagar	Udaipur	The linked lake chain that makes Udaipur the "City of Lakes"
Nakki	Sirohi (Mount Abu)	The highest lake of Rajasthan (about 1,200 m); sacred to the Garasia tribe; legend says the gods dug it with their nails (nakh)
Pushkar	Ajmer	Sacred natural lake ringed by hills with 52 ghats; the only Brahma temple town; Kartik Purnima fair
Ana Sagar	Ajmer	Built by Arnoraja (Anaji), grandfather of Prithviraj Chauhan; Shah Jahan added marble baradaris; Jahangir laid Daulat Bagh beside it
Foy Sagar	Ajmer	Built in 1892 as a famine-relief work under engineer Foy

Lake	District	Fact
Kolayat	Bikaner	Sacred lake of Kapil Muni; Kartik Purnima fair
Siliserh	Alwar	Picturesque lake below the Aravalli, palace on its bank
Gadsisar	Jaisalmer	Rainwater lake built by Rawal Gadsi Singh; gateway built by the courtesan Tilon
Kaylana	Jodhpur	Artificial lake supplying Jodhpur, built in the princely era
Talwara	Hanumangarh	Formed on the Ghaggar; bird habitat
Balsamand	Jodhpur	Old artificial lake (1159 AD) in a garden setting

PYQ PATTERN: Lake–district matching is guaranteed marks: Sambhar–Jaipur/Ajmer edge, Jaisamand–Udaipur, Rajsamand–Rajsamand, Nakki–Sirohi, Kolayat–Bikaner, Gadsisar–Jaisalmer, Siliserh–Alwar, Pushkar–Ajmer.

EXAM TRAP: Jaisamand lies in modern **Salumbar/Udaipur** country (southern Udaipur region), while Rajsamand lake gives the **Rajsamand** district its name — RAS swaps these two Gomati-river lakes.

Climatic Characteristics of Rajasthan

1. Why Rajasthan's climate is like this — five controls

Keep these five reasons in your head and every climate fact below becomes a conclusion instead of a memory load:

- **Latitude:** The Tropic of Cancer crosses the south, so the state is sub-tropical — hot summers are natural.
- **Distance from the sea:** Rajasthan is land-locked in effect; sea winds arrive exhausted, so the west is extremely dry (continental climate — huge day-night and summer-winter temperature gaps).
- **The Aravalli runs parallel to the monsoon:** The Arabian Sea branch blows from the south-west, the same direction the range runs, so the hills fail to lift the winds. The east gets moderate rain; the west stays a desert.
- **Altitude:** The Abu block and the southern hills are cooler — Mount Abu is the state's only hill-station climate.
- **Western disturbances:** In winter, cyclonic winds from the Mediterranean bring light rain called **Mavath** — small in amount, golden for the rabi crop.

Average annual rainfall of the state is about **57.5 cm**, and nearly **90% of it falls in the south-west monsoon season (mid-June to September)**. Rainfall **decreases from the south-east to the north-west** — Kota–Jhalawar are wet, Jaisalmer is the driest corner.

2. Seasons

Season	Months	Character
Summer (Grishma)	March to mid-June	Temperatures 42–48°C in the west; hot wind **loo**; dust storms (**andhi**)
Rainy (Varsha)	Mid-June to September	Monsoon enters via Banswara in the south; most rain from the Arabian Sea branch
Retreating / autumn	October to mid-November	Clearing skies, rising day heat ("October heat"), falling nights
Winter (Sheet)	Mid-November to February	Cold waves from the north-west; frost in Shekhawati and the north; Mavath rain from western disturbances

Temperature extremes the exam loves:

- **India's highest recorded temperature: 51.0°C at Phalodi (Jodhpur region) on 19 May 2016** — a national record from Rajasthan.
- Churu touched **50.8°C (June 2019)**; Churu and Bikaner side are the usual "hottest district" answers in summer.
- The coldest place is **Mount Abu**, where winter nights drop below freezing; frost is common in Sikar–Jhunjhunu–Churu and the northern plains.

3. Rainfall pattern — the map of wet and dry

- **Wettest place: Mount Abu (about 150 cm a year)** — orographic rain on the high Abu block.
- **Wettest district (average): Jhalawar (about 100 cm)**, followed by Banswara (about 92 cm) and Kota.
- **Driest district: Jaisalmer (about 16 cm)**; the **driest place is Sam (Jaisalmer)**.
- The **50 cm isohyet** roughly follows the Aravalli: east of it, rainfall is enough for farming; west of it lies the desert.
- Rainfall is also the most **unreliable (variable)** in the west — Jaisalmer's rain can fail completely in a bad year.
- **Dust storms (andhi)** are most frequent in **Sriganganagar**; rainy days are most numerous in Jhalawar and at Mount Abu.
- **Mavath** (winter rain from western disturbances) is heaviest in the north and north-west and is a blessing for wheat, mustard and gram.

PYQ PATTERN: RAS asks districts in ascending or descending order of average annual rainfall. Anchor your chain with fixed points — Jaisalmer (lowest) at one end and Jhalawar/Abu at the other — then place the middle districts by the rule "rain increases towards the south-east":

Ascending chain (practise it): Jaisalmer → Bikaner → Barmer → Jodhpur → Nagaur → Ajmer → Jaipur → Udaipur → Kota → Banswara → Jhalawar.

EXAM TRAP: Mount Abu is the wettest **place**, Jhalawar the wettest **district**. The exam separates "place" and "district" in the options on purpose.

Approximate normal rainfall of selected districts (cm per year)

District	Rainfall	District	Rainfall
Jaisalmer	about 16	Ajmer	about 52

District	Rainfall	District	Rainfall
Bikaner	about 24	Jaipur	about 56
Barmer	about 28	Udaipur	about 63
Churu	about 35	Kota	about 77
Jodhpur	about 36	Banswara	about 92
Nagaur	about 39	Jhalawar	about 100
Pali	about 49	Mount Abu (place)	about 150

4. Climate zones of Rajasthan

For exam purposes the state is divided into five climatic regions by rainfall:

Zone	Rainfall	Area covered
Arid zone	below 25 cm	Jaisalmer, Bikaner, western Barmer, western Jodhpur
Semi-arid zone	25–50 cm	The Bangar — Nagaur, Sikar, Jhunjhunu, Churu, Pali, Jalore, eastern Jodhpur–Barmer
Sub-humid zone	50–75 cm	Jaipur, Alwar, Ajmer, Bhilwara, Udaipur side
Humid zone	75–100 cm	Kota, Bundi, Banswara, Dungarpur, south-east
Very humid zone	above 100 cm	Jhalawar and the Mount Abu block

In **Koppen's** classification the west is **BWhw** (hot desert), the central belt **BShw** (hot steppe/semi-arid), the east **Cwg** (monsoon with dry winter) and the far south **Aw** (tropical savanna type).

MEMORY HOOK: "BWhw west, BShw middle, Cwg east, Aw south" — one line covers the whole Koppen question.

PYQ PATTERN: Match-the-following on Koppen symbols is common; the desert symbol BWhw attached to Jaisalmer/Bikaner is the most asked pairing.

5. Drought and famine — the climate's dark side

Rajasthan is India's most drought-prone state; some part of it faces drought almost every second year. Local language names the scarcity precisely, and RAS uses these terms:

- **Anna-kal:** famine of grain (crops fail)
- **Jal-kal:** famine of water (drinking water fails)
- **Trin-kal:** famine of fodder (grass fails)
- **Trikal:** all three together — the worst calamity
- The most terrible famine in memory is the **Chhappaniya Akal of 1899–1900** (Vikram Samvat 1956, hence "chhappaniya"), which devastated Marwar and the west

Floods are the opposite, smaller problem: flash floods strike the Luni basin (Barmer–Pali) and the eastern rivers in heavy monsoon bursts, while the Ghaggar sometimes floods Hanumangarh. Drought, not flood, is the state's defining hazard — that is why every other chapter of these notes (irrigation, livestock, drought-hardy crops, water harvesting) exists.

Natural Vegetation, Biodiversity and Conservation

1. The forest picture in one paragraph

Rajasthan is a dry state, so forest cover is thin: recorded forest area is only about **9.5–9.6% of the state's area**, far below the national goal of 33%. What forest exists sits where the rain is — the **south (Udaipur side) and the Aravalli belt**. **Udaipur has the largest forest area of any district**; **Churu has the least**. The southern tribal districts (Pratapgarh, Udaipur, Banswara, Dungarpur) show the highest share of forest within their own area.

Because rainfall rules everything, forest types change in step with the rainfall map: thorn scrub in the desert, mixed dry deciduous in the centre and east, teak in the wet south, and a small patch of sub-tropical hill forest only on Mount Abu.

2. Forest types and where they grow

Forest type	Rainfall belt	Where (districts)	Main species
Tropical thorn forest	below 50 cm	West and north-west — Jodhpur, Barmer, Bikaner, Nagaur, Sikar side	Khejri, rohida, babool, ber, kair, phog
Dry deciduous / mixed deciduous	50–80 cm	Eastern and central plains, Aravalli foothills	Dhok (dhonk), khair, tendu, salar fringe, bamboo patches
Dry teak forest	75–110 cm	South — Banswara, Dungarpur, Udaipur, Chittorgarh, Pratapgarh, Jhalawar, Baran	Teak (sagwan) with tendu, mahua, salar
Salar forest	Aravalli slopes	Udaipur, Rajsamand, Alwar (Sariska), Ajmer side	Salar (Boswellia) dominant on rocky ridges
Dhonk (dhok) forest	50–90 cm	Eastern Aravalli and Hadoti — Karauli, Dholpur, Sawai Madhopur, Kota, Bundi	Dhok (Anogeissus pendula) dominant
Sub-tropical hilly (evergreen) forest	above 110 cm, high altitude	**Mount Abu only (Sirohi)**	Evergreen species with mango, jamun, karonda

PYQ PATTERN: RAS asked directly: "Sub-tropical hilly forests are found in which district?" — answer **Sirohi (Mount Abu)**. It is the ONLY such forest in the state, a tiny green island in the desert state.

PYQ PATTERN: Dry teak = the southern belt (Banswara is the classic answer). Thorn forest = the west (Jaisalmer–Barmer–Bikaner).

The signature trees — learn them as identities

- **Khejri (Prosopis cineraria): State tree** of Rajasthan; the "Kalpavriksha of the desert"; fixes nitrogen, gives sangri pods (the famous panchkuta vegetable). The Bishnoi community protects it — in **1730 at Khejadli (Jodhpur), Amrita Devi Bishnoi and 362 others gave their lives** hugging khejri trees, an event remembered as the world's earliest environment movement and the forerunner of Chipko.
- **Rohida (Tecomella undulata): State flower**; called the "Teak of the desert" (Marwar teak); hard timber of the arid zone.
- **Dhok/Dhonk:** The most widespread tree of the eastern hills; charcoal and fodder.

- **Salar:** Grows on bare Aravalli rocks; yields gum and incense wood.
- **Khair:** Source of kattha (catechu); worked traditionally by the Kathodi tribe.
- **Bamboo:** The south (Banswara, Udaipur) — called "green gold".
- **Tendu:** Leaves for bidi making; Jhalawar, Baran, Chittorgarh, Udaipur.
- **Mahua:** The tribal belt's food-and-flower tree of Banswara–Dungarpur.
- **Sewan grass:** The most nutritious desert grass of Jaisalmer; dhaman, karad and anjan are other valued grasses of the western pastures (called **beed** grasslands).

MEMORY HOOK: Oran — the sacred grove kept by villages for the local deity — is Rajasthan's traditional conservation device; every district has orans where no tree is cut.

3. Protected areas — the conservation map

Rajasthan has **5 national parks, 5 tiger reserves, about 26 wildlife sanctuaries and several conservation reserves**. Learn the national parks and tiger reserves exactly; sanctuaries can be grouped by their star animal.

National parks

National Park	District	Star fact
Ranthambore	Sawai Madhopur	Tiger reserve since 1973 (one of the first nine of Project Tiger); National Park 1980; fort and Padam Talao lake inside; Aravalli meets Vindhya here
Keoladeo (Ghana)	Bharatpur	**Ramsar site (1981) and UNESCO World Heritage Site (1985)**; world-famous bird wetland; water from Ajan Bund (Gambhir)
Sariska	Alwar	Tigers became locally extinct (2004); in **2008 tigers were reintroduced from Ranthambore — the world's first successful tiger reintroduction**
Desert National Park	Jaisalmer, Barmer	Largest national park of the state; home of the **Great Indian Bustard**; Akal Wood Fossil Park inside
Mukundra Hills	Kota, Jhalawar	Newest national park (notified 2012) in the Mukundra range

Tiger reserves (5)

Tiger Reserve	Year / note
Ranthambore	1973 — among India's first nine Project Tiger reserves
Sariska	1978 — the reintroduction reserve
Mukundra Hills	2013 — Hadoti's reserve, receives tigers from Ranthambore landscape
Ramgarh Vishdhari (Bundi)	2022 — India's 52nd tiger reserve; links Ranthambore with Mukundra
Dholpur–Karauli	2023 — the state's 5th tiger reserve, in the Chambal ravine country

PYQ PATTERN: RAS loves the sequence of tiger reserves and the "first reintroduction" fact about Sariska. Ramgarh Vishdhari and Dholpur–Karauli are the two newest — recent papers test them.

EXAM TRAP: Keoladeo is a national park, Ramsar site and World Heritage Site, but it is **not** a tiger reserve.

Important wildlife sanctuaries (group them by specialty)

Sanctuary	District	Known for
Tal Chhapar	Churu	Blackbuck and the special mothiya grass
Kumbhalgarh	Rajsamand, Pali, Udaipur	Leopards and wolves below the great fort
Sitamata	Pratapgarh, Chittorgarh	Flying squirrel and four-horned antelope (chousingha)
Mount Abu	Sirohi	Hill wildlife; green munia bird
Phulwari ki Nal	Udaipur	Dense southern forest corridor
Sajjangarh	Udaipur	One of the state's oldest sanctuaries (1951)
National Chambal Sanctuary	Kota to Dholpur (tri-state)	Gharial, mugger, Gangetic dolphin
Kailadevi	Karauli	Part of the Ranthambore tiger landscape
Jaisamand	Udaipur	Around Jaisamand lake
Jamwa Ramgarh, Nahargarh	Jaipur	Near-city wildlife; Nahargarh has a biological park
Darrah, Jawahar Sagar, Shergarh	Kota, Kota, Baran	Hadoti riverine wildlife
Bassi, Bhainsrorgarh	Chittorgarh	Southern river-and-forest sanctuaries
Todgarh-Raoli	Ajmer, Rajsamand, Pali	Central Aravalli corridor
Van Vihar, Ramsagar	Dholpur	Eastern wetlands

Conservation reserves of note: **Jhalana–Amagarh (Jaipur)** — famous leopard conservation reserve inside a city; **Jorbeer (Bikaner)** — vulture conservation reserve at a carcass dump; **Bisalpur (Tonk)**; **Sundha Mata (Jalore)**; **Gudha Vishnoiyan (Jodhpur)** — Bishnoi-protected blackbuck and chinkara country.

4. Special species and sites

- **Ramsar sites of Rajasthan:** Keoladeo (1981) and Sambhar Lake (1990) are the classic two; **Khichan (Phalodi)** — famous for thousands of wintering demoiselle cranes (kurja) — and **Menar (Udaipur)** were added in 2025 as the state's newest Ramsar sites.
- **Great Indian Bustard (Godawan): State bird**, critically endangered; its last stronghold is the Desert National Park and grasslands like Sorsan (Baran) and Sonkhaliya (Ajmer). Project GIB runs a captive-breeding centre at Sam (Jaisalmer).
- **Lesser florican** breeds at Sorsan and Sonkhaliya grasslands.
- The **Siberian crane** that once wintered at Keoladeo has not been a regular visitor for two decades — if an option claims it is Keoladeo's current star, mark it wrong; the park's identity is its heronries and wintering waterfowl generally.

5. State symbols (geography set)

Symbol	Name
State tree	Khejri
State flower	Rohida
State bird	Godawan (Great Indian Bustard)
State animal (wild)	Chinkara
State animal (domestic)	Camel (declared 2014)
State dance	Ghoomar

MEMORY HOOK: Symbols come in a fixed chain — Tree Khejri, Flower Rohida, Bird Godawan, wild animal Chinkara, domestic animal Camel. Camel is the only symbol with a year attached to it (2014) — a favourite one-mark question.

6. Programmes that fight the desert (development geography)

- **Desert Development Programme (1977-78):** The central programme for the hot arid districts — dune stabilisation, shelter belts, pasture and water harvesting in the Thar blocks
- **Drought Prone Areas Programme:** Its twin, for the semi-arid drought districts of the centre and east
- **Aravalli Afforestation Project:** Ran in the 1990s with Japanese (JICA) support to re-green the degraded Aravalli — the model for later "green wall" thinking along the range
- **Rajasthan Forest Policy 2023:** The current state policy, pushing to expand forest and tree cover and to involve village forest protection committees (the state's joint forest management network)
- **National context:** India hosted the UN desertification conference (UNCCD COP-14, 2019) and pledged land-degradation neutrality — Rajasthan, holding most of India's hot desert, is the frontline state of that pledge

PYQ PATTERN: Programme–purpose matching is standard: Desert Development Programme = hot arid zone only; Aravalli Afforestation = Japanese cooperation; keep the two apart.

Soils of Rajasthan

1. The soil map follows rainfall, west to east

Rajasthan's soils arrange themselves exactly like its rainfall: sand in the west, alluvium in the east, red loam in the south and black soil in the south-east. If you can draw the rainfall map, you already know the soil map.

Soil	Where (districts)	Character and crops
Sandy (desert) soil	Jaisalmer, Barmer, Bikaner, Jodhpur, Churu — the whole Thar	Aridisol/Entisol family; blown sand, very low nitrogen and organic matter, high soluble salts; grows bajra, moth, guar with little water
Brown sandy soil	Nagaur, Sikar, Jhunjhunu, Pali, Jalore — the semi-arid Bangar	Slightly better than pure desert soil; some kankar below

Soil	Where (districts)	Character and crops
Sierozem (grey desert soil)	Nagaur, Ajmer, Pali belt	Pale grey-brown, lime-rich, poor in nitrogen
Alluvial soil	East — Bharatpur, Dholpur, **Jaipur, Dausa, Alwar**, Tonk, Sawai Madhopur	River-borne, deep and fertile; the state's best general farming soil — wheat, mustard, gram, vegetables
Mixed red and yellow soil	Ajmer, Tonk, Sawai Madhopur, Bhilwara, Nagaur	From granite-gneiss; red from iron oxide, yellow where hydrated
Red loam soil	South — Dungarpur, Banswara, Udaipur, Chittorgarh, Pratapgarh	Rich in iron and potash, poor in nitrogen and phosphorus; good for maize, rice, sugarcane
Black (cotton) soil	South-east Hadoti — Kota, Bundi, Baran, Jhalawar	Basalt-born Vertisol; deep, clayey, swells wet and cracks dry; holds moisture; cotton, soybean, coriander, wheat
Saline-alkaline soil (usar/reh)	Depressions of Barmer, Jalore, Nagaur, Bikaner; around salt lakes and the lower Luni tract	White salt crust; needs gypsum and drainage to reclaim
Hill/mountain soil	Aravalli slopes	Thin, stony, poor — only forestry and grazing suit it

PYQ PATTERN: RAS 2021 asked the dominant soil of **Jaipur, Dausa and Alwar** — the answer is **alluvial soil**. In general, whenever a question names eastern districts, think alluvium; southern districts, red loam; Hadoti, black.

EXAM TRAP: "Black soil" belongs ONLY to the Hadoti corner (Kota–Bundi–Baran–Jhalawar) plus small patches near Chittorgarh — not to the whole south. The deep south (Banswara–Dungarpur) is red loam.

2. Soil problems — erosion is the villain

- **Wind erosion** is the state's worst soil problem: summer winds strip the sandy west; stabilising dunes with sewan grass and khejri shelter-belts is the standard remedy (CAZRI, Jodhpur works on this).
- **Water (gully) erosion:** The Chambal and its tributaries have carved the **ravines (beehad/dang)** of Dholpur–Karauli–Sawai Madhopur — India's classic badland. Ravine reclamation (levelling, check dams, plantation) is a long-running programme here.
- **Salinity and waterlogging:** Around salt-lake depressions naturally, and in the Indira Gandhi Canal command (Ganganagar–Hanumangarh) where over-irrigation has raised the water table — the **sem** problem.
- **Desertification:** The desert margin keeps threatening to move east; the Aravalli gaps are the danger corridors, which is why a "green wall" along the Aravalli is discussed.

MEMORY HOOK: One problem per zone — wind in the west, water (ravines) in the east, salt in the depressions, sem in the canal land. Learn problems with their zones, not as a loose list.

Agriculture in Rajasthan

1. The big picture

Farming in Rajasthan is a gamble on one monsoon. About two-thirds of the cropped area is **kharif** (sown June–July), because most of the state has no irrigation. Roughly 40–45% of the net sown area is irrigated; the rest prays for rain. Yet this dry state is a national giant in hardy crops — where India needs a crop that survives heat and thirst, Rajasthan usually leads.

- **Kharif crops:** Bajra, jowar, maize, guar, moth, moong, urad, groundnut, cotton, sesame, soybean, rice (small)
- **Rabi crops:** Wheat, mustard, gram, barley, cumin, coriander, fenugreek, isabgol, taramira
- **Zaid crops:** Watermelon, muskmelon, fodder, vegetables in irrigated pockets

2. Where Rajasthan ranks FIRST in India — the medal table

Crop / item	Rajasthan's rank	Main districts
Bajra (pearl millet)	**1st in production and area** (about 40% of India)	Barmer, Jodhpur, Nagaur, Jaipur, Jalore
Rapeseed–Mustard	**1st** (about 40–48% of India)	Bharatpur, Alwar, Sriganganagar, Tonk, Dholpur
Guar (cluster bean)	**1st** (roughly 70–80% of India)	Bikaner, Jaisalmer, Barmer, Jodhpur, Hanumangarh
Total oilseeds	**1st (among the top)**	Spread over mustard, groundnut, soybean, sesame belts
Gram (chana)	Top two (traditionally 1st)	Bikaner, Jaipur, Ajmer, Nagaur
Moth bean	**1st** (almost the whole national crop)	Bikaner, Churu, Barmer, Jodhpur
Coriander	**1st**	Kota, Baran, Jhalawar (Hadoti)
Fenugreek (methi)	**1st**	Nagaur, Bikaner, Sikar
Isabgol (psyllium)	**1st**	Jalore, Barmer, Jaisalmer
Cumin (jeera)	2nd (after Gujarat)	Jalore, Barmer, Jodhpur, Pali
Wool	**1st**	(see Livestock chapter)
Goats	**1st** in population	(see Livestock chapter)

PYQ PATTERN: RAS asks "Rajasthan is first in India in the production of —" with a group of crops. The safe set is: **bajra, mustard, guar, moth, coriander, fenugreek, isabgol, oilseeds (total), wool**. If "cumin" is the option, remember Gujarat leads cumin; Rajasthan is second.

MEMORY HOOK: Rajasthan's first-rank crops are almost all **dryland rabi/kharif survivors** — mustard, gram, guar, bajra, moth. If it needs little water, Rajasthan probably leads India in it.

Crop belts worth one line each

- **Bajra** is the bread of the desert — largest cropped area of any single crop in the state; Barmer is its biggest home.
- **Wheat** belongs to the canal belt: **Sriganganagar and Hanumangarh** (called the **food basket or granary of Rajasthan**), plus Kota and the eastern plains.
- **Maize** is the crop of the southern tribal belt — Banswara, Dungarpur, Udaipur, Chittorgarh, Bhilwara; this belt keeps Rajasthan among India's top maize states.
- **Cotton:** American (long-staple) cotton in Sriganganagar–Hanumangarh; desi cotton in Banswara–Dungarpur and Jalore.
- **Sugarcane:** Small but present — Sriganganagar, Bundi, Chittorgarh, Udaipur.

- **Groundnut:** Bikaner, Jaipur, Jodhpur are the main pockets.
- **Soybean:** The Hadoti–Malwa edge — Kota, Jhalawar, Baran, Chittorgarh.
- **Spices state:** Rajasthan is a spice bowl — coriander of Hadoti (Ramganj Mandi, Kota, is India's largest coriander market), cumin and isabgol of Jalore–Barmer, methi of Nagaur (Nagauri paan-methi), red chilli of Jodhpur (Mathania), garlic of Baran, seed spices research at Tabiji, Ajmer.
- **Sojat mehndi (henna)** of Pali has a GI tag; opium is grown under licence in Chittorgarh–Pratapgarh.
- **Horticulture:** Kinnow of Sriganganagar; mandarin oranges of Jhalawar ("city of oranges"); pomegranate of Barmer–Jalore; date palm in Bikaner–Jaisalmer (date research farm at Bikaner); ber and aonla in Jaipur–Jobner belt; guava of Sawai Madhopur; mango of Udaipur–Banswara.

EXAM TRAP: For mustard, **Bharatpur** is the emblem district (and Sewar, Bharatpur hosts the national mustard research directorate). For wheat, it is **Ganganagar**. RAS regularly cross-places these two.

3. The ten Agro-Climatic Zones

The state is divided into **ten agro-climatic zones** for research and planning. You do not need every boundary — learn the code, name and one anchor district:

Zone	Name	Anchor districts
I-A	Arid Western Plain	Barmer, Jodhpur
I-B	Irrigated North-Western Plain	Sriganganagar, Hanumangarh
I-C	Hyper-Arid Partially Irrigated Western Plain	Bikaner, Jaisalmer, part of Churu
II-A	Internal Drainage Dry Zone	Nagaur, Sikar, Jhunjhunu
II-B	Transitional Plain of Luni Basin	Jalore, Pali, part of Sirohi
III-A	Semi-Arid Eastern Plain	Jaipur, Ajmer, Tonk, Dausa
III-B	Flood-Prone Eastern Plain	Alwar, Bharatpur, Dholpur, Karauli, Sawai Madhopur
IV-A	Sub-Humid Southern Plain and Aravalli Hills	Udaipur, Rajsamand, Bhilwara, Chittorgarh
IV-B	Humid Southern Plain	Banswara, Dungarpur, Pratapgarh
V	Humid South-Eastern Plateau	Kota, Bundi, Baran, Jhalawar

PYQ PATTERN: Zone–district matching comes straight from this table. The traps are the look-alike zones: I-A (arid, unirrigated) vs I-B (irrigated north-west) vs I-C (hyper-arid) — Ganganagar–Hanumangarh is I-B because of canals, Bikaner–Jaisalmer is I-C.

4. Institutions (one-liners that earn marks)

- **CAZRI (Central Arid Zone Research Institute), Jodhpur (1959):** desert research — dune stabilisation, arid farming, solar and grassland science
- **AFRI (Arid Forest Research Institute), Jodhpur:** arid forestry
- **Agricultural universities:** SKRAU (Bikaner), MPUAT (Udaipur), Agriculture University Jodhpur, SKN Agriculture University (Jobner), Agriculture University Kota
- **National Research Centre on Seed Spices, Tabiji (Ajmer); Directorate of Rapeseed-Mustard Research, Sewar (Bharatpur); Central Institute for Arid**

Horticulture, Bikaner

- The farm-science network (KVKs) and the state seed and warehousing corporations support this structure at district level.

5. Land use in one view

Of the state's reporting area, a little over half is the **net sown area** (roughly 180 lakh hectares); most of the rest is fallow land, culturable waste, grazing land and the thin forest you read about earlier. **Fallow land is unusually large** in Rajasthan because farmers rest rainfed fields in rotation — a dry-year insurance, not laziness. Cropping intensity is modest (about 140%), since a second crop needs irrigation or stored moisture; it is highest in the canal north (Ganganagar–Hanumangarh) and the well-watered east, and lowest in Jaisalmer. Operational holdings are mostly small and marginal, and average holding size keeps shrinking with each agricultural census — together these facts explain why livestock, dairy and wage work must stand beside cropping in the farm economy.

PYQ PATTERN: When a question asks why Rajasthan's cropping intensity is low despite a large sown area, the answer chain is: low irrigation (about 40–45%), high fallows, and rainfall risk — not soil fertility alone.

Livestock of Rajasthan

1. Why livestock matters here

In a state where the rain fails one year in three, animals are the farmer's insurance. Rajasthan holds about **56.8 million livestock (Livestock Census 2019)** — **second in India after Uttar Pradesh**. The state is **1st in India in goat population and in wool production**, **2nd in milk production**, and holds the **bulk (about 85%) of India's camels**. Pastoral communities — Raika/Rabari (camel and sheep), Gurjar and Jat (cattle), Gadaria (sheep) — are built around this wealth.

MEMORY HOOK: "Second in animals, first in goats and wool, second in milk, and almost all of India's camels" — that one sentence is the livestock summary of Rajasthan.

2. Cattle breeds — match the breed to its tract

Breed	Home tract	Identity
Nagauri	Nagaur	Famous draught breed; prized trotting bullocks
Kankrej	Jalore, Barmer (Sanchore side)	Heavy dual-purpose; lyre-shaped horns; also called Sanchori
Tharparkar	Jaisalmer, Barmer	Hardy desert breed from the Thar (Sindh origin); good milker of the dry west
Rathi	Bikaner	Called the "Kamdhenu of the desert"; best milch breed of the arid zone
Gir	Ajmer, Sirohi, Pali side	Origin in Gujarat's Gir; kept across the Aravalli belt
Malvi	Jhalawar (Malwa edge)	Draught breed of the south-east

Breed	Home tract	Identity
Mehwat	Alwar, Bharatpur	Local breed of the Mewat country
Hariana	Eastern districts	From neighbouring Haryana, common in the east

3. Buffalo, sheep, goats

Animal	Breed	Tract / note
Buffalo	Murrah	Alwar, Jaipur belt — the world's best-known buffalo breed, kept widely
Buffalo	Surti	Udaipur, Dungarpur, Banswara (south)
Buffalo	Bhadavari	Dholpur side (Agra region breed)
Sheep	Marwari	Jodhpur, Barmer, Pali — the hardiest and most numerous sheep of the state
Sheep	Malpura	Jaipur, Tonk, Ajmer — heavy, valued for mutton
Sheep	Chokla	Sikar, Jhunjhunu — called the "Indian Merino", finest carpet wool of Rajasthan
Sheep	Jaisalmeri	Jaisalmer — largest-bodied desert sheep
Sheep	Pugal, Magra	Bikaner — Magra gives lustrous carpet wool
Sheep	Sonadi	Udaipur side — called Chanothar locally
Sheep	Nali	Sriganganagar, Hanumangarh (canal tract)
Goat	Sirohi	Sirohi, Udaipur, Ajmer — premier dual-purpose breed
Goat	Marwari	Western districts — hardy meat breed
Goat	Jakhrana	Alwar (Jakhrana village, Behror) — heavy, high-milk breed
Goat	Shekhawati	Sikar, Jhunjhunu — beardless (no wattles) breed
Goat	Sojat	Pali (Sojat) — valued meat breed
Goat	Parbatsari	Nagaur (Parbatsar)
Goat	Barbari	Eastern districts — the "city goat", stall-fed

PYQ PATTERN: Breed–tract matching is a guaranteed format. The classic traps: **Chokla = "Indian Merino"**; **Jakhrana goat = Alwar**; **Rathi cow = Bikaner**; **Kankrej = Sanchore/Jalore**.

EXAM TRAP: The **Malpura is a sheep** and the **Malvi is a cow** — names designed to be swapped in options.

4. The camel — ship of the desert, state animal

- **Breeds: Bikaneri** (the most numerous, heavy baggage breed); **Nachna** of Jaisalmer (Nachna village — the most beautiful and best riding breed); **Jaisalmeri** (riding); **Jalori**, **Kutchi** (Barmer, from Kutch side) and **Gomath** (Phalodi).
- Rajasthan holds about 85% of India's camels; the camel was declared the **state animal (domestic) in 2014** to protect its falling numbers.
- **National Research Centre on Camel, Jorbeer (Bikaner), 1984** — the country's camel headquarters; the International Camel Festival is held at Bikaner.
- Traditional keepers: the **Raika (Rabari)** community.

5. Horses and other animals

- **Marwari horse** of Jodhpur–Marwar is world-famous — inward-turning ear tips, war horse of the Rathores.
- **Malani horse** of Barmer (Malani tract) and **Kathiawari** influence in Jalore–Sirohi complete the horse map.
- Pigs and poultry are minor; backyard poultry is growing in the tribal south.

6. Dairy and animal fairs

- Dairy runs through the **Rajasthan Co-operative Dairy Federation (1977)** and its **Saras** brand; Rajasthan stands **2nd in India's milk production** (after UP), with high per-capita availability.
- The **Central Sheep and Wool Research Institute is at Avikanagar (Tonk)**, where cross-breeds like Avikalin were developed.
- **Animal fairs are geography too** — each fair sits in its breed's home tract: **Nagaur Fair** (cattle, Nagauri bullocks), **Pushkar Fair** (camels, Ajmer), **Tilwara Fair** (Barmer — the Mallinath fair, among the oldest cattle fairs), **Parbatsar Fair** (Nagaur, Tejaji), **Gogamedi Fair** (Hanumangarh), **Jhalrapatan (Chandrabhaga) Fair** (Jhalawar) and **Ramdevra Fair** (Jaisalmer).

PYQ PATTERN: Fair–place matching: Nagaur fair = cattle, Pushkar = camel, Tilwara = Barmer. These three are asked again and again.

Irrigation and Water Resources

1. The water paradox of Rajasthan

Rajasthan has **10.4% of India's land** and about **5.7% of its people**, but only about **1% of the country's surface water**. Irrigation, therefore, decides everything. The broad pattern of sources:

Source	Share of irrigation (approx.)	Where dominant
Wells and tube wells	about 70%	Eastern and central plains — Jaipur, Alwar, Nagaur
Canals	about 28–30%	North (Gang Canal, IGNP, Bhakra) and south-east (Chambal)
Tanks	about 1–2%	South — Bhilwara, Udaipur, Chittorgarh

Sriganganagar is the most irrigated district (canal command); Jaipur leads in well/tube-well irrigation; Bhilwara–Udaipur country leads in tank irrigation.

2. The Indira Gandhi Canal (IGNP) — the lifeline of the desert

- **Source: Harike Barrage** in Punjab, at the confluence of the Sutlej and Beas
- **Structure:** A 204 km feeder canal brings water to the Rajasthan border (entering at Masitawali head, Hanumangarh), then the 445 km main canal runs south-west to **Mohangarh (Jaisalmer)** and the Gadra Road side of Barmer — total system about 649 km
- **History:** Conceived by engineer **Kanwar Sain** (1948 report); its grandfather was Maharaja Ganga Singh of Bikaner, who had already brought Sutlej water by the Gang Canal. Construction began in 1958 as the **Rajasthan Canal**; it was renamed **Indira Gandhi Nahar** in **1984**
- **Command:** Designed for about **19 lakh hectares** in eight districts — Sriganganagar, Hanumangarh, Bikaner, Churu, Jaisalmer, Jodhpur, Barmer, Nagaur. Stage I covers the northern districts; Stage II pushes into Jaisalmer–Barmer–Jodhpur country
- **Lift schemes:** Where the land is higher than the canal, water is lifted — the main ones are the **Kanwar Sain lift (Bikaner–Lunkaransar side)**, **Dr Karni Singh lift (Kolayat)**, **Phalodi lift** and **Pokhran lift**
- It also supplies drinking water to hundreds of desert towns and villages, including Bikaner and Jodhpur cities

MEMORY HOOK: The canal is called the **"Maru Ganga" — the Ganga of the desert**. Questions on source (Harike), entry point (Masitawali, Hanumangarh), renaming (1984) and lift schemes are all standard.

EXAM TRAP: The Indira Gandhi Canal takes water from **Harike on the Sutlej–Beas**, NOT from the Bhakra Dam directly, and NOT from the Ganga.

3. Other canal systems

- **Gang Canal (Old Bikaner Canal):** Built **1923–27 by Maharaja Ganga Singh** of Bikaner, taking Sutlej water from **Hussainiwala (Ferozepur)** into Sriganganagar. It turned Ganganagar into the state's granary and is the oldest modern canal of Rajasthan.
- **Bhakra canals:** Rajasthan's share of Bhakra–Beas waters irrigates Hanumangarh side (Sidhmukh and Nohar projects).
- **Narmada Canal:** Narmada water enters Rajasthan near **Silu (Jalore)** — the canal runs about 74 km in the state, greening Jalore and Barmer villages at the Gujarat border.
- **Gurgaon Canal:** Brings Yamuna water from Okhla to Bharatpur district.
- **Eastern Rajasthan Canal Project (ERCP):** A project to carry surplus Chambal-basin water (via the Parbati–Kalisindh–Chambal link) to **13 districts of eastern and south-eastern Rajasthan**, long demanded as a national project; in 2024 the modified link project was agreed with Madhya Pradesh and renamed the **Ramjal Setu Link Project** by the state.

4. Multipurpose river projects and great dams

Project / Dam	River	District	Identity
Gandhi Sagar	Chambal	(Mandsaur, MP — first dam of the Chambal chain)	First of the four Chambal structures (1960), joint MP–Rajasthan project

Project / Dam	River	District	Identity
Rana Pratap Sagar	Chambal	Chittorgarh (Rawatbhata)	Largest dam reservoir of Rajasthan; atomic power station beside it
Jawahar Sagar	Chambal	Kota	Third Chambal structure (1972)
Kota Barrage	Chambal	Kota	Irrigation barrage (1960); canals serve Kota–Bundi
Bisalpur	**Banas**	Tonk	State's largest drinking-water project; supplies Jaipur, Ajmer, Tonk
Mahi Bajaj Sagar	**Mahi**	Banswara	Joint Rajasthan–Gujarat; named after Jamnalal Bajaj; irrigation + power
Jawai	Jawai (Luni basin)	Pali	Largest dam of western Rajasthan; begun in the princely era by Maharaja Umaid Singh
Jakham	Jakham (Mahi basin)	Pratapgarh	The **highest dam** of Rajasthan (about 81 m)
Meja	Kothari	Bhilwara	Main dam of the Kothari system
Panchna	Gambhir system	Karauli	**Largest earthen (mud) dam** of the state, built across five streams
Ramgarh	Banganga	Jaipur	Old masonry dam near the capital
Som Kamla Amba	Som	Dungarpur	Tribal south's irrigation project
Gudha	Mej	Bundi	Hadoti's storage on a Chambal tributary
Morel	Morel	Sawai Madhopur	On a Banas tributary

PYQ PATTERN: Dam–river matching appears in every second RAS paper: Bisalpur–Banas, Bajaj Sagar–Mahi, Jakham–Jakham (Pratapgarh), Jawai–Jawai (Pali), Panchna–Karauli. Learn these five pairs cold.

PYQ PATTERN: Superlatives asked: **tallest dam = Jakham**, **largest earthen dam = Panchna**, **largest reservoir = Rana Pratap Sagar**, **largest western-Rajasthan dam = Jawai**.

5. Traditional water wisdom — the desert's own technology

The exam treats these as geography, and they are pure Rajasthan:

- **Khadin:** A runoff-farming system of **Jaisalmer** (developed by the Paliwal Brahmins centuries ago) — a bunded low land catches monsoon runoff; crops grow on the stored moisture after the water sinks.
- **Johad:** Small earthen check-dams of the **Alwar** country. Revived from 1985 by **Rajendra Singh (Tarun Bharat Sangh)** — the dried **Arvari river flowed again**, making him the "Waterman of India".

- **Nadi:** Village pond of the western districts, the basic drinking-water source.
- **Tanka:** Underground cistern in houses and forts (Bikaner, Jaisalmer) storing roof rainwater.
- **Beri:** Shallow wells in the Jaisalmer dune country tapping perched water.
- **Baori/Bawdi (stepwell): Chand Baori of Abhaneri (Dausa)** is among India's deepest and largest stepwells; Raniji ki Baori is Bundi's pride.
- **Jhalara:** Rectangular stepped tanks of Jodhpur.
- The state ran the **Mukhyamantri Jal Swavlamban Abhiyan (2016)** to revive such traditional structures village by village.

MEMORY HOOK: Water structures match their districts like breeds match tracts — Khadin–Jaisalmer, Johad–Alwar, Chand Baori–Dausa (Abhaneri), Jhalara–Jodhpur.

6. Shared rivers — Rajasthan's water agreements in brief

Almost every big river of the state is shared, so its water is divided by agreement; these three arrangements cover most questions:

- **Ravi–Beas waters:** Rajasthan holds an **8.6 MAF share** of the surplus Ravi–Beas flows (1955 allocation, reaffirmed in later agreements) — this Punjab water is what the Indira Gandhi Canal carries into the desert
- **Chambal waters:** Shared **equally (50:50) with Madhya Pradesh** under the Chambal Valley Project agreements
- **Mahi waters:** Shared with **Gujarat** — Mahi Bajaj Sagar (Rajasthan) and Kadana (Gujarat) work as one system

EXAM TRAP: The Sutlej–Beas water for IGNP comes under the **Ravi–Beas** arrangement, even though the canal physically lifts water at **Harike** below the Sutlej–Beas confluence. Keep the two facts in separate boxes.

Population of Rajasthan (Census 2011)

Census 2011 is the examination standard for all population questions. Anchor the state picture first, then the district extremes — RAS almost always asks extremes and orders.

1. The state picture — anchor figures

Indicator (Census 2011)	Rajasthan	India (for contrast)
Total population	6,85,48,437 (6.85 crore) — 5.66% of India	121 crore
Decadal growth 2001–11	**21.3%**	17.7%
Population density	**200 per sq km**	382 per sq km
Sex ratio	**928** females per 1,000 males	943
Child sex ratio (0–6 years)	**888**	919
Literacy rate	**66.1%** (Male 79.2%, Female 52.1%)	74.0%
Rural / Urban share	75.1% / 24.9%	68.9% / 31.1%
Scheduled Caste share	17.8%	—
Scheduled Tribe share	13.5%	—

Indicator (Census 2011)	Rajasthan	India (for contrast)
Population aged 0–6	15.5% of total	—

Reading the table as a story: Rajasthan grows faster than India (21.3%), is half as crowded (200 vs 382), and its weakest points are exactly the three gender-and-education numbers — sex ratio 928, child sex ratio 888 (it FELL from 909 in 2001) and female literacy 52.1%. These contrasts are the examiner's hunting ground.

MEMORY HOOK: "200 – 928 – 888 – 66.1" — density, sex ratio, child sex ratio, literacy. Four numbers carry most of this chapter.

2. District extremes — population and density

Fact	Highest	Lowest
Population	**Jaipur (66.26 lakh)** — then Jodhpur, Alwar, Nagaur, Udaipur	**Jaisalmer (6.70 lakh)** — then Pratapgarh (8.68 lakh)
Density	**Jaipur (595)**	**Jaisalmer (17)**
Decadal growth	**Barmer (32.5%)**	**Sriganganagar (10.0%)**

Density order at the top (descending): **Jaipur 595 → Bharatpur 503 → Dausa 476 → Alwar 438 → Dholpur 398**. At the bottom: **Jaisalmer 17 → Bikaner 78 → Barmer 92 → Churu 147 → Jodhpur 161**.

The logic is simple and worth stating: density follows water. The canal-irrigated north and the eastern plains are crowded; the empty Thar holds 17 persons per sq km in Jaisalmer — less than one-thirtieth of Jaipur.

3. Sex ratio and child sex ratio — the south leads, the east lags

Indicator	Highest districts	Lowest districts
Sex ratio (overall)	**Dungarpur 994**, Rajsamand 990, Pali 987, Pratapgarh 983, Banswara 980	**Dholpur 846**, Jaisalmer 852, Karauli 861, Bharatpur 880
Child sex ratio (0–6)	**Banswara 934**, Pratapgarh 933	**Jhunjhunu 837**, Sikar 848

Learn the pattern, not just the names: the **tribal south (Banswara, Dungarpur, Pratapgarh)** has the healthiest sex ratios in the state — tribal society values daughters more equally — while the **eastern Jat-belt districts (Dholpur, Karauli, Bharatpur)** and prosperous **Shekhawati (Jhunjhunu, Sikar)** record the worst child sex ratios.

PYQ PATTERN: RAS 2023/2024 pattern — districts are given and you must arrange sex ratio or child sex ratio in descending order. Anchor pairs: Dungarpur top (994), Dholpur bottom (846); Banswara top in child sex ratio (934), Jhunjhunu bottom (837). Everything else falls between these anchors.

EXAM TRAP: **Highest sex ratio = Dungarpur**, but **highest child sex ratio = Banswara**. Swapping the two southern districts is the standard trap.

4. Literacy — Kota at the top, Jalore at the bottom

Descending order of literacy rate (Census 2011) — a directly repeated PYQ format:

Rank	District	Literacy
1	**Kota**	76.6%
2	Jaipur	75.5%
3	Jhunjhunu	74.1%
4	Sikar	71.9%
5	Alwar	70.7%
...last	Barmer	56.5%
...last	Banswara	56.3%
...last	Pratapgarh	56.0%
...last	Sirohi	55.3%
Last	**Jalore**	**54.9%**

PYQ PATTERN: The standard question is: "Arrange in descending order of literacy — Kota, Jaipur, Jhunjhunu, Sikar." The order above is the answer; Shekhawati (Jhunjhunu, Sikar) punches above its weight because of its education tradition and army-recruitment culture, while the desert west (Jalore, Sirohi, Barmer) and the tribal south (Banswara, Pratapgarh) lag at the bottom for opposite reasons — remoteness in one, poverty in the other.

EXAM TRAP: Kota is first in **overall** literacy (76.6%) and also first in **female literacy** (about 65.9%); Jalore is last in both (female literacy about 38.5%). Do not place Jaipur first — it is second.

Rural literacy is 61.4% and urban literacy 79.7%; the female literacy jump from 43.9% (2001) to 52.1% (2011) is the state's one big education gain of the decade.

5. Where the SC and ST populations concentrate

The two social maps are mirror images, and questions exploit the contrast:

- **Scheduled Castes (17.8% of the state):** Highest share in the canal-irrigated north — **Sriganganagar (about 36.4%, the state's highest)** and Hanumangarh (about 27.8%) — where farm labour settled along the canals. The lowest SC shares are in the tribal south (Dungarpur, about 3.8%).
- **Scheduled Tribes (13.5% of the state):** Highest in the southern hills — **Banswara (76.4%)** — and negligible in the middle-west (Nagaur records the lowest share, about 0.3%).
- The two maps rarely overlap: the north is SC-heavy and ST-light; the south is ST-heavy and SC-light.

EXAM TRAP: "Highest SC population share district" = Sriganganagar; "highest ST share district" = Banswara. The options deliberately pair them — one belongs to the canal north, the other to the tribal south.

6. Migration — the moving population

Population geography is incomplete without movement. Three flows define Rajasthan:

- **Seasonal labour migration from the tribal south:** Every dry season, large numbers of Bhil workers from Banswara, Dungarpur and Udaipur travel to Gujarat's cities, farms and construction sites (and to Madhya Pradesh), returning with the monsoon. Low local irrigation and tiny holdings push this flow.

- **The historic Marwari diaspora:** Merchants from Marwar and Shekhawati migrated over two centuries to Kolkata, Mumbai, Chennai and the North-East, becoming India's most famous trading community — "Marwari" is a business identity across India, rooted in the desert towns of this state.
- **In-migration to the growth poles:** Jaipur, Kota (students), Bhiwadi–Neemrana (industry), the canal colonies of Ganganagar and the mining belts draw workers from within Rajasthan and from UP, Bihar and West Bengal.

MEMORY HOOK: Push from the south (poverty, no irrigation), pull to the north-east corridor (Delhi–NCR industry) and the capital — migration in Rajasthan flows from the wet-poor hills to the dry-rich plains, the reverse of the rainfall map.

Urbanization in Rajasthan

1. A still-rural state

Census 2011 counted **1.70 crore urban people — 24.9% of the state** (against India's 31.1%). Rajasthan urbanises slowly because its industrial base is thin and its desert districts can support few towns. There were **297 towns** in the state in 2011.

- **Most urbanised district: Kota (60.3% urban)** — industry, coaching and the Chambal command
- **Second: Jaipur (52.4% urban)**
- **Least urbanised districts: Dungarpur (6.4%) and Banswara (7.1%)** — the tribal south remains almost entirely rural

2. The million-plus cities — exactly three in 2011

City	Population 2011	Identity
Jaipur	about 30.5 lakh (largest)	Capital; among India's ten largest cities; "Pink City"
Jodhpur	about 10.3 lakh (city), 11.4 lakh urban agglomeration	Marwar's capital; second city of the state
Kota	about 10.0 lakh — just over the million line	Chambal city; industry and the national coaching hub

Below them stand Bikaner (about 6.4 lakh), Ajmer (about 5.5 lakh), Udaipur (about 4.5 lakh), Bharatpur and Alwar. The state's Class-I cities (above one lakh) in 2011 also included Bhilwara — the textile city nicknamed the "Manchester of Rajasthan" — Pali, Sriganganagar, Tonk, Sikar, Hanumangarh and Beawar; each of these towns grew around one economic engine: textiles, canals, trade or industry. The urban growth rate of the decade (about 29%) ran fastest exactly here, in the industrialising towns, while old pilgrim and administrative towns grew slowly.

Notice the urban logic of the state: every large town sits either on the eastern fertile corridor (Jaipur–Alwar–Bharatpur), on a canal command (Ganganagar, Bikaner, Kota), or in the Mewar mining-industrial belt (Udaipur, Bhilwara). The western desert holds only oasis towns — Jaisalmer, Barmer, Jodhpur — spaced far apart on old trade routes. **Bhiwadi** (in the Alwar region, inside the National Capital Region) is the state's fastest-growing industrial town, with Neemrana's industrial and Japanese zone nearby. Alwar and Bharatpur districts form Rajasthan's share of the **NCR**.

PYQ PATTERN: "How many million-plus cities did Rajasthan have in Census 2011?" — **Three: Jaipur, Jodhpur, Kota.** The trap options add Udaipur or Bikaner, which had NOT crossed ten lakh in 2011.

Jaipur, Udaipur, Ajmer and Kota were selected under the national **Smart Cities Mission** — the state's four smart cities.

MEMORY HOOK: Urban Rajasthan in one line — "a quarter urban, three million cities, Kota most urbanised, Dungarpur least."

Tribes of Rajasthan

1. The tribal map

Scheduled Tribes number **92.4 lakh — 13.5% of the state's population**; Rajasthan stands **fourth among Indian states in tribal population**. Twelve communities are listed as Scheduled Tribes: **Bhil, Bhil-Meena, Damor, Dhanka, Garasia, Kathodi, Kokna, Koli Dhor, Meena, Naikda, Patelia and Sahariya**.

The tribes are concentrated in the **southern hill belt** — the Vagad and Mewar hills where forest and rainfall support them:

District	Share of ST in district population
Banswara	**76.4% — highest in the state**
Dungarpur	70.8%
Pratapgarh	63.4%
Udaipur	49.7%
Sirohi, Chittorgarh	substantial pockets

The Tribal Sub-Plan (TSP) area covers this southern belt — Banswara, Dungarpur and Pratapgarh fully, and parts of Udaipur, Sirohi, Chittorgarh, Rajsamand and Pali.

2. The main tribes — one identity each

Meena — the most populous tribe

- **Largest tribe of Rajasthan by population (about 43 lakh)**
- Country: the **east** — Jaipur, Dausa, Sawai Madhopur, Karauli, Bharatpur, Alwar (the old Matsya–Dhundhar country)
- Settled agriculturists; the Meenas ruled the Amber–Dhundhar country before the Kachhwahas, and revere their descent from the Matsya (fish) incarnation of Vishnu; two traditional divisions — **Ujjala and Maila**

Bhil — the second largest, and the largest tribe of India

- About 41 lakh in Rajasthan; the Bhil are the **largest tribal community of India as a whole**
- Country: the **south** — Udaipur, Banswara, Dungarpur, Chittorgarh, Rajsamand
- The name comes from "bil" — the bow; they are born archers. Their hamlets (**phala/pal**) scatter over hill slopes; the village headman is the **Gameti**
- History remembers **Rana Punja Bhil**, the Bhil chief who stood with Maharana Pratap at Haldighati; their spring festival is **Bhagoriya** (the haat festival before Holi in the Banswara belt)

EXAM TRAP: "Largest tribe of Rajasthan" is the classic two-way trap: by Census 2011 population, **Meena is first and Bhil second** in Rajasthan; but the **Bhil is the largest tribe of India**. Read whether the question asks about Rajasthan or India before answering.

Garasia — third largest

- Country: **Sirohi (Abu Road side), Udaipur and Pali**, around the Abu hills; divided into Rajput Garasia and Bhil Garasia
- Their **Valar dance** is performed without instruments; the **Ghotia Amba fair** is their great gathering; Nakki lake at Abu is sacred to them

Sahariya — the only PVTG of Rajasthan

- Rajasthan's only **Particularly Vulnerable Tribal Group (PVTG)**
- Country: **Baran district — the Shahbad and Kishanganj blocks** of Hadoti, almost nowhere else
- A forest-dependent community of gatherers and shifting labour; their hamlet is called a **Saharana**; the **Sitabari fair (Baran)** is their great festival (the largest tribal fair of Hadoti)

Damor, Kanjar, Kathodi and others

- **Damor:** Dungarpur (Simlawada side) and Banswara; a small community of Gujarat origin
- **Kanjar:** Hadoti belt — Kota, Bundi, Baran, Jhalawar; famous for the women's **Chakri dance**
- **Kathodi:** Udaipur (Jhadol, Kotra); originally from Maharashtra; traditional makers of kattha from khair wood
- **Dhanka, Naikda, Patelia, Kokna, Koli Dhor, Bhil-Meena:** Smaller communities of the southern and eastern belts

3. Tribal institutions and memory points

- **Govind Guru** led the Bhil awakening of the south; his followers were massacred at **Mangarh (Banswara) in 1913** — about 1,500 tribals killed, called the "Jallianwala Bagh of the tribals". Mangarh Dham is now a national memorial, and **Govind Guru Tribal University at Banswara (2012)** carries his name.
- **Beneshwar Dham (Dungarpur)** — at the **confluence (Triveni) of the Som, Mahi and Jakham** rivers — hosts the great tribal fair on **Magh Purnima**, called the **Kumbh of the tribals**; it is the seat of Saint Mavji's tradition.
- **Eklavya Model Residential Schools** and the tribal research institute at Udaipur (Manikya Lal Verma Tribal Research Institute) serve this belt.

PYQ PATTERN: Three facts repeat endlessly: (1) **Sahariya = only PVTG, Baran**; (2) **highest ST share district = Banswara**; (3) **Beneshwar = Som–Mahi–Jakham sangam**. Collect all three as free marks.

Minerals of Rajasthan — Metallic and Non-Metallic

1. The museum of minerals

Rajasthan is called the **"Museum of Minerals"** — more than **80 types of minerals** are found here, and the state mines more varieties than any other Indian state. Its strength is non-metallic and

non-ferrous wealth: it leads India in most industrial minerals and in lead-zinc-silver, and its Aravalli belt is one of the world's great storehouses of base metals. The Department of Mines and Geology has its headquarters at **Udaipur**, and the public-sector miner is **RSMML (Rajasthan State Mines and Minerals Ltd, 1974)**.

Near-monopoly facts to fix first — Rajasthan produces:

- **100% (sole producer) of India's wollastonite and jasper**
- Practically **all of India's lead and zinc** (the only lead-zinc producing state of consequence) and the **bulk of its silver**
- About **90% of India's marble**, about **70% of its sandstone**
- Over **80% of India's gypsum**, about **three-fourths of its rock phosphate**
- The **leading share** of feldspar, fuller's earth, ball clay, steatite and gem garnet

2. The master mineral–district table (the PYQ battlefield)

RAS asks mine–district matching in almost every paper. Practise this table as pairs:

Mineral	District / mine	Note
Zinc–Lead	**Rampura Agucha (Bhilwara)**	Among the world's largest zinc-lead mines; Hindustan Zinc
Zinc–Lead, Silver	**Sindesar Khurd (Rajsamand)**	Among the world's largest silver-producing mines
Zinc–Lead	Rajpura Dariba (Rajsamand), Kayad (Ajmer)	HZL mines of the Aravalli belt
Zinc–Lead (ancient)	**Zawar (Udaipur)**	Among the world's oldest zinc mines — medieval zinc distillation was practised here
Copper	**Khetri–Singhana (Jhunjhunu)**	Hindustan Copper; Kolihan and Chandmari mines; India's classic copper belt
Copper	Kho-Dariba (Alwar)	Second copper centre
Iron ore	**Morija–Banol, Neemla–Raisalo (Jaipur–Dausa belt)**	Hematite-magnetite; also Nathra-ki-Pal (Udaipur), Dabla (Sikar side), Pur–Banera (Bhilwara)
Manganese	Banswara (Lilwani, Kalakhunta)	Small but examinable deposits of the south
Tungsten	**Degana (Nagaur)**	India's only tungsten-producing centre (wolframite); also Balda (Sirohi)
Rock phosphate	**Jhamarkotra (Udaipur)**	India's largest rock-phosphate deposit; RSMML works it
Marble	**Makrana (Nagaur)**	White marble of Taj Mahal fame (GI-tagged); Kishangarh (Ajmer) is Asia's largest marble market/mandi
Marble	Rajsamand	Leading marble-producing district by volume; green marble at Keshariaji (Udaipur)
Granite	**Jalore**	"Granite City of Rajasthan"; also Sirohi, Barmer

Mineral	District / mine	Note
Gypsum	**Jamsar (Bikaner)**	Largest gypsum mine area; also Bhadwasi (Nagaur), Barmer
Lignite	**Palana, Barsingsar (Bikaner)**; Kapurdi–Jalipa, Giral (Barmer); Soneri (Nagaur)	Palana is the oldest lignite field; feeds Barsingsar (NLC) and Barmer lignite power plants
Petroleum and gas	**Barmer basin**; gas in Jaisalmer basin	Mangala–Bhagyam–Aishwariya fields (2004 Mangala discovery — India's biggest onshore find of its time); the HPCL–state refinery at Pachpadra (Balotra)
Wollastonite	Sirohi (Bel ka Magra), Udaipur, Ajmer	Sole producing state
Jasper	Jodhpur, Bikaner, Jaisalmer	Sole producing state
Feldspar	Ajmer	Leading producer; pegmatite belt
Mica	Bhilwara, Tonk, Ajmer	The Bhilwara belt is the classic mica tract
Asbestos	Udaipur, Rajsamand, Bhilwara, Dungarpur	Chrysotile/amphibole belt of the Aravalli
Emerald	**Kaliguman (Rajsamand)**, Rajgarh (Ajmer)	Rajasthan's emerald (panna) country
Garnet	Rajmahal (Tonk), Sarwar (Ajmer)	Gem garnet near-monopoly
Fuller's earth (Multani mitti)	Barmer, Bikaner, Jaisalmer	Largest producer
Ball clay, china clay	Bikaner (Kolayat side), Nagaur	Ceramic minerals
Limestone	Jaisalmer (Sonu — steel-grade), Chittorgarh–Nimbahera belt, Nagaur, Kota	Chittorgarh is the cement hub; Rajasthan among India's top cement producers
Salt	Sambhar, Didwana, Pachpadra	From the salt lakes (see Lakes chapter)
Fluorite	Mando-ki-Pal (Dungarpur)	Known fluorite centre
Uranium	Rohil (Sikar)	Deposit identified by atomic minerals exploration
Gold	Bhukia–Jagpura (Banswara)	Exploration-stage gold field of the south
Potash	Nagaur–Ganganagar basin	Huge evaporite reserves under exploration

PYQ PATTERN: The four pairings that return most often — **Rampura Agucha–zinc (Bhilwara)**, **Khetri–copper (Jhunjhunu)**, **Degana–tungsten (Nagaur)**, **Jhamarkotra–rock phosphate (Udaipur)**. Then the second ring: Makrana–marble, Jamsar–gypsum, Zawar–oldest zinc, Mangala–oil.

EXAM TRAP: **Zawar is in Udaipur district**, not Rajsamand — its near neighbours Rajpura Dariba and Sindesar Khurd ARE in Rajsamand. The three HZL centres of the south are deliberately juggled in options.

3. Energy minerals — one connected story

The west holds the fuels. **Lignite** (brown coal) lies under Bikaner (Palana, among India's oldest lignite mines; Barsingsar feeds an NLC power plant) and Barmer (Kapurdi–Jalipa and Giral feed the

Barmer lignite plant). **Petroleum** transformed Barmer after the **Mangala discovery of 2004** — with Bhagyam and Aishwariya it made Rajasthan a major crude producer, and a **9 MMTPA refinery-cum-petrochemical complex at Pachpadra** (HPCL Rajasthan Refinery, a joint venture with the state) anchors downstream industry. **Natural gas** comes from the Jaisalmer basin fields (Manohar Tibba, Tanot, Dandewala side) feeding the Ramgarh gas power plant.

4. Rajasthan Mineral Policy 2024 — highlights

The state replaced its 2015 policy with the **Rajasthan Mineral Policy 2024**. For the exam, hold these points:

- Target: raise the mining sector's share in state GSDP from about **2.3% to 5% by 2029-30** (and towards 8% by 2046-47)
- Push to auction more mineral blocks with **pre-embedded clearances** to cut delays
- **"One District One Mineral"** promotion for local mineral-based industry
- Promotion of **M-Sand** (manufactured sand) to relieve river-sand pressure
- Zero-waste and scientific mining, stronger action on illegal mining, and better use of District Mineral Foundation (DMF) funds for mining-affected people
- The parallel **Minor Mineral** rules govern the vast sandstone, marble, granite and masonry-stone sector that employs much of the Aravalli and Jodhpur country

PYQ PATTERN: Recent papers test policy facts — the year (2024), the GSDP-share target (5% by 2029-30) and the One District One Mineral idea are the likely one-liners.

5. Mineral-based industry — where the rocks become factories

Minerals earn their real money in processing, and the industry map mirrors the mineral map:

Industry	Centre	Raw material link
Zinc smelters	Debari (Udaipur), Chanderiya (Chittorgarh) — Hindustan Zinc	Zawar, Rajpura Dariba, Sindesar Khurd concentrates
Lead smelter and silver refining	Chanderiya (Chittorgarh)	Same lead-zinc chain — silver is recovered here
Copper smelter	Khetri (Jhunjhunu) — Hindustan Copper	Khetri–Singhana ore
Cement cluster	Chittorgarh–Nimbahera, Beawar, Kotputli, Lakheri (Bundi), Jaisalmer side	Limestone belts; Chittorgarh is called the cement capital of the state
Fertilizers	Gadepan (Kota) — Chambal Fertilisers; phosphatic units in Udaipur–Kota country	Rock phosphate of Jhamarkotra, gas and Chambal water
Marble processing and trading	Kishangarh (Ajmer), Makrana, Rajsamand	Makrana and Rajsamand quarries
Granite cutting and polishing	Jalore, Barmer, Sirohi	Jalore granite
Glass and ceramics	Jaipur–Sanganer side, Bikaner	Silica sand, ball clay and china clay of Bikaner–Nagaur
Salt works and chemicals	Sambhar, Didwana, Pachpadra	Salt-lake brines
Textile-linked mineral use	Pali (dyeing) draws on the Bandi river — an industry-environment conflict the papers sometimes cite	—

MEMORY HOOK: See the chain as a flow: Aravalli ore → Udaipur–Chittorgarh smelters; Vindhyan limestone → Chittorgarh–Kota factories; desert brines → salt and soda country. Every industrial centre in the table sits on top of its raw material.

Tourism Geography of Rajasthan

Tourism in Rajasthan is geography wearing a crown: forts sit on Aravalli spurs, palaces rise from Mewar's lakes, desert tourism rides on the Thar's dunes, and the state's fairs are pinned to lakes, rivers and breed tracts you have already studied. RAS keeps the tourism questions geographical — circuits, UNESCO sites, locations and settings.

1. The tourist circuits

The state organises tourism into circuits. Learn the core list with one anchor each:

Circuit	Anchor places	Identity
Dhundhar	Jaipur, Dausa, Samode	The capital region; part of the Golden Triangle (Delhi–Agra–Jaipur)
Mewar	Udaipur, Chittorgarh, Rajsamand, Nathdwara	Lakes and great forts
Marwar	Jodhpur, Pali, Nagaur	Mehrangarh and the desert gateway
Shekhawati	Sikar, Jhunjhunu (Mandawa, Nawalgarh)	Painted havelis — an "open-air art gallery"
Hadoti	Kota, Bundi, Baran, Jhalawar	Palaces, stepwells, the Chambal
Vagad	Banswara, Dungarpur	Tribal south, Mahi country
Merwara	Ajmer, Pushkar, Kishangarh	Dargah, Brahma's Pushkar, marble mandi
Alwar	Alwar, Sariska, Siliserh	Tiger country nearest Delhi
Bharatpur (Braj edge)	Bharatpur, Deeg, Dholpur	Keoladeo and the Jat palaces
Desert	Jaisalmer, Bikaner, Barmer side	Dunes, camel safaris, border tourism
Godwad	Mount Abu, Ranakpur, Jalore side	The hill-and-temple south-west
Ranthambore	Sawai Madhopur	Tiger tourism

2. UNESCO World Heritage Sites — the protected list

Rajasthan holds four entries on the World Heritage List:

Site	Year	Note
Keoladeo National Park	1985	Natural site — the Bharatpur wetland
Jantar Mantar, Jaipur	2010	Sawai Jai Singh II's stone observatory — the largest of his five observatories, with the world's largest stone sundial
Hill Forts of Rajasthan (serial, 6 forts)	2013	**Chittorgarh, Kumbhalgarh, Ranthambore, Gagron, Amber, Jaisalmer**
Jaipur City (walled city)	2019	The planned pink city of 1727

PYQ PATTERN: The serial "Hill Forts" list is the favourite — six forts, and candidates forget **Gagron**. Remember Gagron separately: it is the only **water fort (jal durg)** of the six, standing at the confluence of the Ahu and Kali Sindh in Jhalawar, with no foundation laid on the rock beneath.

3. Forts read as geography

- **Kumbhalgarh (Rajsamand):** Built by Rana Kumbha; its wall runs about **36 km — among the longest walls in the world** after the Great Wall of China; birthplace of Maharana Pratap; guards the Aravalli crossing into Marwar.
- **Chittorgarh:** India's **largest fort by area**, on a long mesa above the Berach river; Vijay Stambha (1448, nine storeys) celebrates Rana Kumbha's victory.
- **Jaisalmer Fort (Sonar Quila):** Built 1156 AD by Rawal Jaisal on the Trikuta hill; yellow sandstone glows golden; one of the world's few **living forts** — thousands still reside inside.
- **Mehrangarh (Jodhpur):** Rao Jodha's 1459 fort on the steep Chidiya Toonk hill, rising sheer above the blue city.
- **Junagarh (Bikaner):** Built by Raja Rai Singh (1589–94) — unusually, a great fort **on the plain, not on a hill**.
- **Amber (Jaipur):** Man Singh I's palace-fort above Maota lake; Sheesh Mahal inside.
- **Lohagarh (Bharatpur):** Surajmal's "Iron Fort", which repelled Lord Lake's attacks in 1805.
- **Bhatner (Hanumangarh):** Among the oldest forts of the state, standing sentinel on the Ghaggar plain at the desert's northern gate.
- **Taragarh twins:** Ajmer's Taragarh (the "Garh Beetli") and Bundi's Taragarh share a name — a ready-made trap.

4. Places with a geographical tag

- **Mount Abu (Sirohi):** The state's **only hill station** — Guru Shikhar, Nakki lake, Dilwara marble temples, sunset point; receives the state's highest rainfall.
- **Udaipur:** "City of Lakes", "Venice of the East" — Pichola, Fateh Sagar, Jaisamand country.
- **Pushkar (Ajmer):** Sacred lake town, the camel fair on Kartik Purnima — among the world's largest camel fairs.
- **Sam dunes (Jaisalmer):** The face of desert tourism; Desert Festival (February) and camel safaris.
- **Jhalana (Jaipur):** Urban leopard safari — wildlife tourism inside a capital city.
- **Abhaneri (Dausa):** Chand Baori stepwell and Harshat Mata temple.
- Heritage trains: the **Palace on Wheels (1982)** luxury train and the **Fairy Queen** — among the world's oldest working steam locomotives (1855), running the Delhi–Alwar heritage route.

5. Fairs and festivals pinned to the map

Fair / Festival	Place	Season / note
Pushkar Fair	Pushkar, Ajmer	Kartik Purnima; camels and cattle
Desert Festival	Jaisalmer	February; dune culture
Camel Festival	Bikaner	January
Nagaur Fair	Nagaur	Jan–Feb; great cattle fair
Marwar Festival	Jodhpur	October

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Fair / Festival	Place	Season / note
Mewar Festival	Udaipur	Spring (March–April)
Kumbhalgarh Festival	Rajsamand	December
Beneshwar Fair	Dungarpur (Mahi sangam)	Magh Purnima; tribal Kumbh
Kolayat Fair	Bikaner	Kartik Purnima; Kapil Muni
Urs	Ajmer Dargah	Khwaja Moinuddin Chishti
Teej and Gangaur	Jaipur (grandest), Udaipur	Monsoon and spring women's festivals
Khatu Shyamji Fair	Sikar	Phalguna
Gogamedi Fair	Hanumangarh	Bhadrapada
Ramdevra Fair	Jaisalmer (Runicha)	Bhadrapada

Facts for the margin: Rajasthan was the **first state to give tourism the status of an industry (1989)**; the Rajasthan Tourism Development Corporation (RTDC) dates to 1978; the Tourism Policy of 2020 refreshed incentives; and Jaipur's seat in the **Golden Triangle** with Delhi and Agra brings the state its heaviest international footfall.

MEMORY HOOK: Close the chapter with the UNESCO chain — "Birds (Keoladeo 85), Stars (Jantar Mantar 2010), Forts (2013), City (Jaipur 2019)."

6. Pilgrimage and border tourism — the living circuits

- **Nathdwara (Rajsamand):** Shrinathji's temple — the devotional capital of the Pushtimarga, drawing pilgrims to the Banas valley town near Udaipur
- **Ajmer–Pushkar pair:** The Sufi Dargah of Khwaja Moinuddin Chishti and Brahma's Pushkar stand barely 11 km apart across the Nag Pahar — a shared pilgrim circuit unique in India
- **Khatu Shyamji (Sikar)** and **Salasar Balaji (Churu):** Shekhawati's twin pilgrimage magnets, drawing lakhs on foot in Phalguna
- **Karni Mata, Deshnok (Bikaner):** The famous rat temple on the Bikaner–Nagaur road
- **Ramdevra/Runicha (Jaisalmer)** and **Gogamedi (Hanumangarh):** Folk-deity centres (Ramdevji, Gogaji) that pull devotees across the northern desert in Bhadrapada
- **Border tourism in the Thar:** Tanot Mata temple and the **Longewala** war memorial (Jaisalmer) — the 1971 battlefield — along with munificent desert posts near Sriganganagar, form a patriotic circuit at the Pakistan frontier
- **Kuldhara (Jaisalmer):** The abandoned Paliwal village in the dunes — heritage tourism built on a desert legend

PYQ PATTERN: Pilgrim-centre matching is gentle but regular: Nathdwara–Shrinathji, Deshnok–Karni Mata, Khatu–Shyamji, Salasar–Balaji, Ramdevra–Ramdevji, Gogamedi–Gogaji. Match the deity to the town and district together.

Rapid Revision — Superlatives and One-Line Facts

Use this closing chapter in the last week before the paper. Every line is a potential one-mark question.

1. District superlatives (Census 2011 frame)

Superlative	Answer
Largest district by area	Jaisalmer (38,401 sq km)
Smallest district by area	Dholpur (3,034 sq km)
Most populous district	Jaipur
Least populous district	Jaisalmer
Highest density	Jaipur (595)
Lowest density	Jaisalmer (17)
Highest sex ratio	Dungarpur (994)
Lowest sex ratio	Dholpur (846)
Highest child sex ratio	Banswara (934)
Lowest child sex ratio	Jhunjhunu (837)
Highest literacy	Kota (76.6%)
Lowest literacy	Jalore (54.9%)
Highest decadal growth	Barmer (32.5%)
Lowest decadal growth	Sriganganagar (10.0%)
Highest ST share in population	Banswara (76.4%)
Most urbanised district	Kota (60.3%)
Least urbanised district	Dungarpur (6.4%)
Largest forest area	Udaipur
Least forest area	Churu
Wettest district	Jhalawar
Driest district	Jaisalmer
Most irrigated district	Sriganganagar
District touching most districts	Pali (8)

2. Records and firsts

- Highest peak: **Guru Shikhar, 1,722 m** (Sirohi)
- Oldest fold mountains of India: **Aravalli**
- Longest river (connected with the state): **Chambal, 1,051 km**; longest river entirely within the state: **Banas, 512 km**
- Shortest river: **Kakni** (about 17 km, Jaisalmer)
- Only perennial river: **Chambal**
- Only river crossing the Tropic of Cancer twice: **Mahi**
- Largest inland salt lake of India: **Sambhar**
- Highest lake: **Nakki** (Mount Abu)
- Tallest dam: **Jakham** (Pratapgarh); largest earthen dam: **Panchna** (Karauli); largest reservoir: **Rana Pratap Sagar**; largest dam of western Rajasthan: **Jawai** (Pali)
- Hottest recorded temperature in India: **51.0°C, Phalodi (2016)**
- Wettest place: **Mount Abu** (about 150 cm); driest place: **Sam** (Jaisalmer)
- First tiger reserve of the state: **Ranthambore (1973)**; newest: **Dholpur–Karauli (2023)**

- First successful tiger reintroduction in the world: **Sariska (2008)**
- Only PVTG tribe: **Sahariya (Baran)**
- Most populous tribe: **Meena**; largest tribe of India found here: **Bhil**
- World's oldest zinc mines: **Zawar (Udaipur)**; among the world's largest zinc mines: **Rampura Agucha (Bhilwara)**
- India's only tungsten centre: **Degana (Nagaur)**; largest rock-phosphate deposit: **Jhamarkotra (Udaipur)**
- State's only hill station: **Mount Abu**; state's only river port of history: none — Rajasthan is landlocked in effect
- First state to give industry status to tourism: **Rajasthan (1989)**
- Longest canal system of the state: **Indira Gandhi Canal (about 649 km)**, from **Harike**
- Longest wall of a fort in India: **Kumbhalgarh (about 36 km)**

3. The ten sentences that summarise Rajasthan

- Sand in the west (61%), stone in the middle (9%), soil in the east (23%), black plateau in the south-east (7%) — the four divisions.
- The Aravalli runs parallel to the monsoon, so the west is a desert and the east is a plain.
- Chambal is the only perennial river; Banas is the longest home river; Luni turns salty after Balotra; Mahi cuts the Tropic twice.
- Sambhar is India's salt bowl; Mewar holds the sweet lakes.
- Abu is the wettest place, Jaisalmer's Sam the driest; Jhalawar is the wettest district.
- Thorn forest in the west, teak in the south, evergreen only on Abu.
- Black soil only in Hadoti; red loam in the south; alluvium in the east.
- If the crop is hardy — bajra, mustard, guar, moth — Rajasthan leads India.
- Goats, wool and milk put the state at the top of livestock India; the camel is its signature.
- 200 – 928 – 888 – 66.1: density, sex ratio, child sex ratio, literacy — the four numbers of Census 2011.

MEMORY HOOK: Revise in this order — map first (divisions), then water (rivers, lakes, canals), then people (census, tribes), then wealth (soil, crops, minerals). RAS papers are built in exactly this order too.

StuDy